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Evolution of open quantum systems under continuous measurement of their environment

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« La gloire de Dieu, c'est de cacher les choses; La gloire des rois, c'est de sonder les choses. »
— Proverbes 25:2

CONTENTS

CONTEXT AND MOTIVATIONS

This document is the report of an internship carried out as part of a Master's degree in fundamental and numerical physics (CompuPhys). My internship has been conducted within the team of Theorical Physics — *équipe de Physique Théorique* ΦTh — of the *UTINAM Institute* in the building of the *Observatory of Besançon*.

This work aims to study, from an academical perspective at first, the subject of continuous measurement considering open quantum systems which was a different topic from my supervisor's specific area of research. Our primary motivation was therefore initially to educate ourselves on a field of quantum physics that is well known to a large part of the scientific community. By familiarizing ourselves with the topic, we decided to adopt a particular frame of report due to our journey of understanding. Therefore, the structure of this document is meant to help discovering and exploring the fundamentals of quantum continuous measurement to any unfamiliar reader. We wanted to make it as illustrative and pedagogical as possible, thus, simple examples will be used to graphically depict mathematical and physical concepts in the chapter concerning continuously monitored quantum open systems.

Introduction

"Quantum mechanics is the description of the behavior of matter and light in all details and, in particular, of the happenings on an atomic scale" said *R.Feynman* in one of his lectures [1]. In this framework, position is no longer a single well-defined value prior to measurement, instead, position is described by a probability density all across space. Consequently, the notion of a trajectory as a continuous succession of points in spacetime is lost in quantum dynamics. The definition of trajectory will therefore be addressed, but one question remains paramount when mentioning systems: how can we even define a quantum system properly?

In the ideal case, systems are treated as closed, *i.e.*, isolated from any external environment and interacting with nothing at all. This modelling allows quantum mechanics to describe dynamics of systems with unitary and deterministic terms (the state at time t is determined solely by knowing the state at time t_0) using the Schrödinger equation. [2]
However in the physical world, systems do "feel" their surroundings, *i.e.*, they interact with their environment. This interaction should be taken into account to obtain a more complete and physically relevant representation. In doing so, we describe the so-called *Open Quantum System* (OQS) theory [3].

Other useful theories have been developed through the last decades, allowing quantum mechanics' description to be more and more complete. One of them being the Quantum Measurement Theory (QMT) [4], where the concept of measurement is fully described. Measuring a system has a tremendous impact on the its quantum properties. Furthermore, since any experimental probe inevitably interacts with the system, it is crucial to figure out the nature of this interaction. Measuring a system without impacting its quantum properties leads to the concept of weak measurements, that we will develop further below.

By studying QMT, one can thus notice that measuring a quantum system alters it. The measurement usually interacts so strongly with the system that it collapses into a state — an eigenstate of the observable — that lost interesting quantum properties. For example it can lose its entanglement[1]. Along the same lines, quantum systems can be in superposition of states, taking *Schrödinger's cat* thought experiment, this corresponds formally to a superposition of 'alive' and 'dead' states prior to measurement. Measuring the state of the cat will determine its fate, alive or dead, it is then "stuck" in the state measured and cannot be written as a superposition again. The idea of **weak measurements** is to **measure the environment** coupled with the system. Instead of altering the state, we obtain pieces of information on the system. The measurements can even be conducted continuously, maximizing the amount of data the observer can get. These are the global ideas behind **Quantum Continuous Measurements** (QCM) [5].

Practically, this work will be organized as follows: the first section covers an academical progression from basics in quantum mechanics up to the aforementioned theories. A second section will cover the photodetection and homodyne detection of open quantum systems. The qubit and harmonic oscillator will be used to illustrate our point. The final part is the study of a system of interest on which we apply quantum continuous measurement. (Si on a le temps.)

[1] $|\psi\rangle_{AB} = \frac{|00\rangle+|11\rangle}{\sqrt{2}} \xrightarrow[\text{in A}]{\text{measuring 0}} |00\rangle_{AB} = |0\rangle_A \otimes |0\rangle_B$ a product state (thus non-entangled).

Chapter I

Basics of quantum mechanics

I.1 Generalities on closed quantum systems

I.1.1 States and time evolution

In quantum mechanics a physical system's state is described by a vector in a Hilbert space:

$$|\psi\rangle \in \mathcal{H}. \tag{I.1}$$

The Hamiltonian $\hat{H}$ is a specific Hermitian[2] operator representing classically the total energy of a system. This relationship will be further detailed in Section I.1.2. Physically, the evolution of a system is governed by its energy, the *Schrödinger equation* states it explicitly:

$$i\hbar \frac{\mathrm{d}\,|\psi(t)\rangle}{\mathrm{d}t} = \hat{H}(t)\,|\psi(t)\rangle\,. \tag{I.2}$$

The solution of the Schrödinger equation can be expressed as an unitary[3] operator $\hat{U}$ such that,

$$|\psi(t)\rangle = \hat{U}(t,t_0)\,|\psi(t_0)\rangle\,, \tag{I.3}$$

with initial condition $U(t_0,t_0) = \mathbb{I}$. By injecting the previous relation in the state equation, we get

$$\frac{\mathrm{d}\hat{U}}{\mathrm{d}t}(t,t_0) = -\frac{i}{\hbar}\hat{H}(t)\hat{U}(t,t_0). \tag{I.4}$$

We will simply assume the Hamiltonian to be time-independent in the following, nevertheless, solutions exist in the time-dependent case [3]. For us, the equation of evolution, solution of the Schrödinger equation, is

$$|\psi(t)\rangle = \underbrace{e^{-\frac{i\hat{H}(t-t_0)}{\hbar}}}_{\hat{U}(t,t_0)}\,|\psi(t_0)\rangle\,. \tag{I.5}$$

This solution is indeed **unitary** and we can use $\hat{U}^\dagger(t,t_0)$ on the left of both sides of the Eq. (I.5) to obtain $|\psi(t_0)\rangle = \hat{U}^\dagger(t,t_0)\,|\psi(t)\rangle$, i.e., we described **reversible** dynamics.

I.1.2 Observables

In quantum mechanics, any physical quantity is represented by an **observable** which is an operator fulfilling two properties: the eigenvectors of the observable form an orthonormal

[2] $\hat{H}^\dagger = \hat{H}$, where † denotes the conjugate transpose, defining the Hermitian adjoint.
[3] $\hat{U}\hat{U}^\dagger = \hat{U}^\dagger\hat{U} = \mathbb{I}$, where $\mathbb{I}$ is the identity.

basis of the Hilbert space $\mathcal{H}$, and, the corresponding eigenvalues are necessarily real [2]. In finite-dimensional Hilbert spaces, the definition of Hermitian operator[4] is sufficient for it to be an observable. In the case of infinite dimension Hilbert spaces, the definition of an observable requires more rigorous mathematical foundations (self-adjoint operators, the details of which lie beyond the purview of this study). In this sense, we only consider finite dimension cases in the following. We thus define the interval $I_{\mathcal{H}} = [\![1; \dim(\mathcal{H})]\!]$ throughout this document, also, we lighten the notation by omitting $\hat{\bullet}$ symbols on operators hereafter.

A natural operator whose eigenbasis $\{|\phi_n\rangle\}_{n\in I_{\mathcal{H}}}$ is particularly significant is the Hamiltonian H, as its eigenvalues $\{E_n\}_{n\in I_{\mathcal{H}}}$ correspond to the allowed energy values of a system. We can understand the relation between an operator, its eigenvectors and eigenvalues with its corresponding eigenequation:

$$\forall n \in I_{\mathcal{H}}, \quad H|\phi_n\rangle = E_n |\phi_n\rangle . \tag{I.6}$$

Since $\{|\phi_n\rangle\}_{n\in I_{\mathcal{H}}}$ forms a basis of the Hilbert space, one can describe any state $|\psi\rangle$ in this very basis:

$$\forall |\psi\rangle \in \mathcal{H}, \qquad |\psi\rangle = \sum_{n\in I_{\mathcal{H}}} c_n |\phi_n\rangle , \tag{I.7}$$

where $c_n = \langle\phi_n|\psi\rangle$. This kind of description in a base is obviously valid for any observable, not only for the Hamiltonian.

Finally, we mention the necessity of normalizing states, which is important when calculating probabilities (the Section I.2 makes it obvious). In order to normalize a state, one needs to set $\|\psi\| = \sqrt{\langle\psi|\psi\rangle} = 1$, which is equivalent to the condition $\langle\psi|\psi\rangle = \sum_n |c_n|^2 = 1$.

I.1.3 Purity of states

The concept of *pure state* arises from the necessity to write the state of a system in a different way than that used so far in this document. Indeed, certain ways of preparing a system — such as a statistical mixture[5]— require another formalism because such processes prevent the state of the system from being expressed as a single $|\psi\rangle$ (or as a superposition of state vectors). To take into account such cases, we introduce the density matrix hereafter.

A density matrix ρ is an operator of the Hilbert space $\mathcal{H}$ (and also an operator of the dual space $\mathcal{H}^*$), which follows a set of properties:

- ρ is Hermitian: $\rho = \rho^\dagger$, implying it can be diagonalized and necessarily with a positive eigenvalues, $\mathrm{Sp}(\rho) \subset \mathbb{R}^+$.
- $\mathrm{Tr}[\rho] = 1$ necessarily.
- ρ is positive semi-definite: $\forall |\psi\rangle \in \mathcal{H}, \quad \langle\psi|\rho|\psi\rangle \geq 0$.

The density matrix formalism is particularly useful because it generalizes Dirac formalism. Indeed, if a state can be written as $|\psi\rangle$ then $\rho = |\psi\rangle\langle\psi|$ and we call it a **pure state**. Otherwise, the state is non-pure and can only be expressed with a density matrix. Also, pure states satisfy stronger properties than those defining a general density matrix. Specifically, being idempotent $\rho^2 = \rho$ by construction, implying that $\mathrm{Tr}\left[\rho^2\right] = \mathrm{Tr}[\rho] = 1$ in the case of pure states.

[4]The eigenvectors of a Hermitian operator always span the entire Hilbert space (Spectral Theorem). Using the Gram-Schmidt process, we can always find a complete orthonormal basis from the operator's eigenspaces.

[5]In the Appendix A, we explain how to prepare a system as a statistical mixture or as a superposition of states. However, we recommend to read the Section I.2 on quantum measurement theory before.

I.1.4 Postulates of quantum mechanics

Physicists developed a corpus of the main properties of quantum mechanics, here are summarized the so-called postulates of quantum mechanics [2]:

1. The state of a system is represented by a density matrix $\rho \in \mathrm{End}(\mathcal{H})$ respecting properties I.1.3. For pure states we can describe them $|\psi\rangle \in \mathcal{H}$, or, by using the pure matrix density formalism $\rho_p = |\psi\rangle\langle\psi|$.
2. Physical quantities $\mathcal{A}$ are associated with operators $\hat{A}$, called observables. In the context of finite dimension space, they are simply Hermitian operators.
3. The only possible outcomes of a measurement of $\mathcal{A}$ are the eigenvalues a of $\hat{A}$, defined by $\hat{A}\,|a\rangle = a\,|a\rangle$.
4. When measuring $\mathcal{A}$ on a normalized state, we know the probability to get one of the possible eigenvalues: $\mathbb{P}[a_n]$ (which depends on the spectrum of the observable).
5. If the measurement of the physical quantity $\mathcal{A}$ on the system gives the result a_n, the state of the system immediately after the measurement collapses into the eigenstate $|\phi_n\rangle$.
6. The time evolution of the wavefunction $|\psi(t)\rangle$ is governed by the *Schrödinger* equation:
$$\frac{\mathrm{d}\,|\psi(t)\rangle}{\mathrm{d}t} = -\frac{i}{\hbar} H(t)\,|\psi(t)\rangle\,. \tag{I.8}$$
For density matrices, the evolution is driven by the *Liouville-von Neumann* equation:
$$\frac{\mathrm{d}\rho}{\mathrm{d}t}(t) = -\frac{i}{\hbar}\left[H(t), \rho(t)\right], \tag{I.9}$$
where $[\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A}$ is called the commutator of $\hat{A}$ and $\hat{B}$.

I.1.5 Multipartite systems

In quantum mechanics a system can be composed of multiple subsystems, we then call the global composition a multipartite system. The density matrix introduced in the previous sections is a useful tool to describe the state of such system and subsystems. If we consider an Hilbert space $\mathcal{H}_{\mathcal{AB}}$ composed of two physical systems A and B: $\mathcal{H}_{\mathcal{AB}} = \mathcal{H}_A \otimes \mathcal{H}_B$. Note that this implies $\dim(\mathcal{H}_{AB}) = \dim(\mathcal{H}_A) \times \dim(\mathcal{H}_B)$. The total system state is therefore represented by a density matrix $\rho_{AB} \in \mathcal{H}_A \otimes \mathcal{H}_B$. The expression of the density matrix representing the state of the subsystem A is defined by
$$\rho_A = \mathrm{Tr}_B[\rho_{AB}], \tag{I.10}$$
and is called a *reduced density matrix*. Moreover, we defined in this equation a new function called a *partial trace*. The idea behind the notation is to get rid of the B components in the tensor product of the total state ρ_{AB}. Explicitly, we can express the mean value of an operator of the subsystem A in the total space. Let $\hat{O} = \hat{O}_A \otimes \mathbb{I}_B$, and specify the bases of A and B: namely $\{|\phi_a\rangle\}_a$ and $\{|\phi_b\rangle\}_b$. The average of the operator $\hat{O}$ is expressed as

$$\begin{aligned}\left\langle \hat{O} \right\rangle = \mathrm{Tr}\left[\rho_{AB}\hat{O}\right] &= \sum_{a=1}^{\dim(\mathcal{H}_A)} \sum_{b=1}^{\dim(\mathcal{H}_B)} \left(\langle\phi_a| \otimes \langle\phi_b|\right)\rho_{AB}\left(\hat{O}_A \otimes \mathbb{I}_B\right)\left(|\phi_b\rangle \otimes |\phi_a\rangle\right) \\ &= \sum_{a=1}^{\dim(\mathcal{H}_A)} \langle\phi_a| \underbrace{\left(\sum_{b=1}^{\dim(\mathcal{H}_B)} \langle\phi_b|\,\rho_{AB}\,|\phi_b\rangle\right)}_{\mathrm{Tr}_B[\rho_{AB}]:=\rho_A} \hat{O}_A\,|\phi_a\rangle = \mathrm{Tr}_A\left[\rho_A\hat{O}_A\right].\end{aligned} \tag{I.11}$$

Therefore, for any local observable of A, the expectation value is encoded by the reduced density matrix ρ_A. Note that this average of a local observable does not require explicit knowledge on the total state, while the average still takes into account the correlations between A and B (due to $\rho_A = \mathrm{Tr}_B[\rho_{AB}]$).

The bipartite results shown above can be generalized to multipartite systems. However, in the following we will consider a system S coupled to an environnement E only, which is thus bipartite.

I.2 Quantum Measurement Theory (QMT)

In the QMT [6, 7], measuring consists in applying a projector to a state, so, we call this process a *projective measurement*. The set of projective measurements is called a Projection-Valued Measure (PVM) [8]. For the sake of simplicity, we work — as until now — with non-degenerate spectra for the observables. Since an observable has a set of eigenvectors $\{|a_n\rangle\}_{n\in I_{\mathcal{H}}}$, we can build a projector corresponding to one of them:

$$P_{a_n} = |a_n\rangle\langle a_n| \,. \tag{I.12}$$

This definition implies four important properties:

$$P_{a_n}^\dagger = P_{a_n}, \qquad P_{a_n}^2 = P_{a_n}, \qquad P_{a_n}P_{a_m} = P_{a_n}\delta_{nm}, \qquad \sum_n P_{a_n} = \mathbb{I}.$$

To represent the quantum mechanics definition of a measurement, the projective measurement needs to fulfill some requirements. The first requirement is to generate the probability of observing the result of a measurement on a system. Thus, the probability to observe a_n is defined by

$$\mathbb{P}_{|\psi\rangle}[a_n] = \langle\psi|P_{a_n}|\psi\rangle = |\langle\psi|a_n\rangle|^2, \qquad \mathbb{P}_\rho[a_n] = \mathrm{Tr}\left[\rho P_{a_n}\right], \tag{I.13}$$

where the subscripts $|\psi\rangle$ and ρ refer respectively to the representation of pure states or to density matrix more general formalism.

$$\overline{|\psi_{a_n}\rangle} = P_{a_n}|\psi\rangle\,, \qquad \overline{\rho_{a_n}} = P_{a_n}\rho P_{a_n}. \tag{I.14}$$

Here the $\overline{\bullet}$ symbol means that the state is not normalized, which will be an important notation for later, see Section II.5. By normalizing we find the proper expression of a state after a measurement with result a_n:

$$|\psi_{a_n}\rangle = \frac{P_{a_n}|\psi\rangle}{\sqrt{\langle\psi|P_{a_n}|\psi\rangle}}, \qquad \rho_{a_n} = \frac{P_{a_n}\rho P_{a_n}}{\mathrm{Tr}\left[\rho P_{a_n}\right]}. \tag{I.15}$$

The obtained state is said **conditioned by the measurement** result a_n, hence the importance of the index a_n on states obtained after measurement. We point out that the normalization factor is exactly the probability to observe the result a_n for the density matrix (or the square root of the probability, when using ket formalism). In the end, projective measurement fulfill the properties of quantum mechanics by construction because they generate probabilities (respecting the fact they need to be positive and their sum is 1), and can also be used to write the expression of collapsed states after measurement.

However, the definition of PVM has been found too restrictive to represent every kind of measurement allowed by quantum mechanics [9, 10]. One can define a more general kind of measurement, that fulfills quantum mechanics needs while relaxing some properties: the orthogonality between each measure elements is not required to generate probabilities. Therefore, one can build the so-called Positive Operator-Valued Measure (POVM), implicitly defined by the probability to observe the result a_n:

$$\mathbb{P}_\psi[a_n] = \langle\psi|\Pi_{a_n}|\psi\rangle\,, \qquad \mathbb{P}_\rho[a_n] = \mathrm{Tr}\left[\rho\Pi_{a_n}\right], \tag{I.16}$$

where the operator Π_{a_n} is called the *effect* [7] for the result a. Moreover, the set of all effects $\{\Pi_{a_n}\}_{n\in I_{\mathcal{H}}}$ forms the POVM. As stated, the only properties we force the POVM to respect are:

$$\text{positive semi-definite: } \forall\,|\phi\rangle \in \mathcal{H},\ \langle\phi|\Pi_{a_n}|\phi\rangle \geq 0, \qquad \sum_{n\in I_{\mathcal{H}}} \Pi_{a_n} = \mathbb{I}.$$

These properties imply that POVM generate proper probabilities. Furthermore, the (normalized) states obtained after finding the result a_n are now

$$|\psi_{a_n}\rangle = \frac{K_{a_n}|\psi\rangle}{\sqrt{\langle\psi|\Pi_{a_n}|\psi\rangle}}, \qquad \rho_{a_n} = \frac{K_{a_n}\rho K_{a_n}^\dagger}{\mathrm{Tr}\left[\rho\Pi_{a_n}\right]}, \tag{I.17}$$

where we introduced the so-called *Kraus* operator K_{a_n}, which respects $\Pi_{a_n} = K_{a_n}^\dagger K_{a_n}$.

POVM can describe processes of quantum measurement where the results are ignored, just as in the thought experiment for statistical mixture of states developed in Appendix A. In such a case, we can describe statistically the state without knowing the measurement results:

$$\rho = \sum_{n\in I_{\mathcal{H}}} \mathbb{P}[a_n]\rho_{a_n} = \sum_{n\in I_{\mathcal{H}}} K_{a_n}\rho K_{a_n}^\dagger. \tag{I.18}$$

The corresponding relationship for PVM is obtained by replacing both K_{a_n} and $K_{a_n}^\dagger$ by P_{a_n}.

Finally, one can observe that POVM generalize PVM in the sense that projectors are special cases of POVM elements (adding self-adjoint property and orthogonality to POVM gives a PVM). Furthermore, PVMs make states collapse on eigenstates that are necessarily orthogonal to each other, which is not necessarily the case for POVMs. A brief example of POVM forcing a state to collapse onto "non-orthogonal states" is when considering initially the state $|\psi\rangle = \frac{|g\rangle+|e\rangle}{\sqrt{2}} \equiv |+\rangle$. If we choose the Kraus operators $K_0 = |g\rangle\langle g|$ and $K_1 = |+\rangle\langle e|$, which fulfill $K_0^\dagger K_0 + K_1^\dagger K_1 = \mathbb{I}$, and, that form the POVM $\{|g\rangle\langle g|\,, |e\rangle\langle e|\}$. Then, when the result of the measurement is 0, we find that the state collapses to $|\psi_0\rangle = |g\rangle$, and when finding 1 it collapses to $|\psi_1\rangle = |+\rangle$. However, $\langle\psi_0|\psi_1\rangle = \frac{1}{\sqrt{2}} \neq 0$, thus the states on which the initial state can collapse are not orthogonal to each other.

I.3 Open Quantum Systems (OQS)

In the previous part, we saw that one can fully describe a system, observables, states and their evolution. However, this description applies to the ideal case where the quantum system is considered isolated from any external interaction. Instead, in the OQS theory systems interact locally with an environment, which itself interacts with a wider environment, etc... To avoid considering infinite chains of interaction, the theory truncates the environment to a local vicinity of the system. Thus, we approximate the total system T as a closed quantum system, within which the quantum system of interest S is called an open quantum system. Moreover, as explained S couples with its environment E, thus giving rise to a dynamical regime distinct from that of closed quantum systems.

I.3.1 Building the formalism

We will present here a derivation of the Lindblad master equation using the *input-output formalism* [5, 11]. This derivation can alternatively be carried out using a microscopic approach [3]. Since a full derivation of the input-output formalism is mathematically challenging, we will only provide its key equations and their physical meaning.

As seen in Section I.1.5, the Hilbert space of the total system is divided into $\mathcal{H}_T = \mathcal{H}_S \otimes \mathcal{H}_E$. We consider as initial state a factorized state: $\rho_T = \rho_S \otimes \rho_E$. From now on we will use a typographic convention where $\rho_T \equiv \boldsymbol{\varrho}$ and $\rho_S \equiv \rho$, such that the last assumption rewrites $\boldsymbol{\varrho} = \rho \otimes \rho_E$. The system Hamiltonian H_S depends on which physical system is studied and is left free. Instead, the environment is chosen as a collection of quantum harmonic oscillators, the environment Hamiltonian being thus defined by

$$H_E = \int_0^{+\infty} \hbar\omega b_\omega^\dagger b_\omega \mathrm{d}\omega, \tag{I.19}$$

where the operators of the environment satisfy $\left[b_\omega, b_{\omega'}^\dagger\right] = \delta(\omega - \omega')$. This Hamiltonian represents a continuum of bosonic modes of frequency $\omega \in \mathbb{R}^+$, where each mode is represented by a quantum harmonic oscillator. We define the so-called interaction Hamiltonian by

$$H_I = i\hbar \int_0^{+\infty} \sqrt{\frac{k(\omega)}{2\pi}} \left(c b_\omega^\dagger - c^\dagger b_\omega\right) \mathrm{d}\omega, \tag{I.20}$$

where c is an operator of the system, and $\omega \mapsto k(\omega)$ is a function representing the coupling strength between S and E for each considered mode at frequency ω. This coupling is treated as a perturbation: the interaction has a smaller impact on the dynamics than the free evolution. This form of interaction Hamiltonian can be found, for example, in the case of a two-level system interacting with a radiation field via a dipole interaction, once the rotating wave approximation has been performed. Consequently by using Eq. (I.19,I.20), we define the total Hamiltonian:

$$H_T = \overbrace{H_S \otimes \mathbb{I}_E + \mathbb{I}_S \otimes H_E}^{H_0} + H_I. \tag{I.21}$$

By switching to the interaction picture (detailed in Appendix B), we can get rid of the free dynamics governed by H_0. Indeed, in the interaction picture the Liouville-von Neumann equation becomes

$$\frac{\mathrm{d}\tilde{\boldsymbol{\varrho}}}{\mathrm{d}t} = -\frac{i}{\hbar}\left[\tilde{H}_I, \tilde{\boldsymbol{\varrho}}\right], \tag{I.22}$$

in which we used the following notation: for any operator $O(t)$ we denote by $\tilde{O}(t)$ its expression in the interaction picture, such that $\tilde{O}(t) = U_0^\dagger(t,t_0) O(t) U_0(t,t_0)$. The unitary operator U_0 is the free evolution operator generated by H_0, and is defined as $U_0(t,t_0) = e^{-\frac{i}{\hbar}H_0(t-t_0)}$.

Before calculating explicitly the expression of $\tilde{H}_I$, we need to make some assumptions which are listed below.

- The *first Markovian assumption* we consider the interaction strength to be constant in a (large) bandwidth $\mathcal{W} = [\omega_0 - \theta; \omega_0 + \theta]$ (ω_0 is the natural frequency of S), so that $\forall \omega \in \mathcal{W}, \quad k(\omega) = k$. The coupling strength is considered negligible outside of $\mathcal{W}$.

- The system operator acquires a time-dependent phase in the interaction picture: $\tilde{c}(t) = c e^{-i\omega_0(t-t_0)}$. This ansatz is motivated by the explicit form of operators in the interaction picture when considering elementary systems [5].

- We assume that $k \ll \omega_0$, which establishes $\tau = \frac{1}{k}$ as the dominant timescale for the dynamics.

We can now express the interaction Hamiltonian in the interaction picture by

$$\tilde{H}_I(t) = i\hbar\sqrt{k}\left[cb_t^\dagger - c^\dagger b_t\right], \tag{I.23}$$

in which we defined the input field operator b_t as

$$b_t = \frac{1}{\sqrt{2\pi}} \int_{\mathcal{W}} b_\omega e^{-i(\omega-\omega_0)(t-t_0)} \mathrm{d}\omega. \tag{I.24}$$

To characterize the environmental operators b_t, we can compute their canonical commutation relation:

$$\left[b_t, b_{t'}^\dagger\right] = \frac{1}{\sqrt{2\pi}} \int_{-\theta}^{\theta} e^{-i\omega(t-t')} \mathrm{d}\omega = \delta(t-t'). \tag{I.25}$$

To obtain Eq. (I.25), we assumed the frequency bandwidth $\mathcal{W}$ to be much larger than all other frequencies of the dynamics, implying in particular $k \ll \theta$. In this limit, the integration in Eq. (I.25) can be extended to the entire real line [5, 11, 12], which yields the definition of the Dirac delta distribution. Note that by defining $\mathcal{W} = [\omega_0 - \theta; \omega_0 + \theta]$, we implicitly state that $\omega_0 \geq \theta$. This condition gives again the third assumption mentioned earlier: $k \ll \theta \leq \omega_0 \Rightarrow k \ll \omega_0$. The delta-distribution we just obtained in Eq. (I.25) is not satisfying because the commutator becomes singular at equal times. We will now open a brief parenthesis on input operators and Wiener increments to bypass this singularity.

We define a Wiener process in quantum stochastic calculus as

$$W(t, t_0) = \int_{t_0}^{t} b_{t'} \mathrm{d}t'. \tag{I.26}$$

We can further derive a differential quantity using calculus [13], called a quantum Wiener increment and defined as

$$\mathrm{d}W_t = \lim_{\delta t \to \mathrm{d}t} \left[W(t+\delta t, t_0) - W(t, t_0)\right] = b_t \mathrm{d}t. \tag{I.27}$$

Using Eq. (I.26), the quantum Wiener increment canonical commutation relation (at same times) is

$$\left[\mathrm{d}W_t, \mathrm{d}W_t^\dagger\right] = \mathrm{d}t. \tag{I.28}$$

However, by calculating directly the same commutator with the right hand side of Eq. (I.27), we find

$$\left[\mathrm{d}W_t, \mathrm{d}W_t^\dagger\right] = \left[b_t, b_t^\dagger\right] \mathrm{d}t^2. \tag{I.29}$$

When we combine this result with Eq. (I.28), we get

$$\left[b_t, b_t^\dagger\right] \mathrm{d}t = \mathbb{I}. \tag{I.30}$$

If we now define a set of bosonic operators B_t such that $B_t = b_t\sqrt{\mathrm{d}t}$, then

$$\left[B_t, B_t^\dagger\right] = \mathbb{I}, \tag{I.31}$$

which is the canonical commutation relation of creation/annihilation bosonic operators[6]. Since $\{B_t\}_{t\in\mathbb{R}}$ forms an infinite set of proper annihilation operators, the environment can be modeled

[6]In the input-output formalism, the annihilation operator acts on its Fock basis as $B_t\left|n\right\rangle_t = \sqrt{n}\left|n-1\right\rangle_t$ (with the vacuum condition $B_t\left|0\right\rangle_t = 0$), whereas the adjoint (creation operator) acts as $B_t\left|n\right\rangle_t = \sqrt{n+1}\left|n+1\right\rangle_t$.

as a continuous collection of quantum harmonic oscillators. In this framework, at any time t, the system couples to the corresponding temporal bosonic mode defined by B_t. After evolving over the infinitesimal duration $\mathrm{d}t$, the system decouples from the mode at t and interacts with the next mode at $t + \mathrm{d}t$.

We now make use of one more assumption to derive the evolution of the state $\rho(t)$ of the open quantum system S.

- The *Born-Markov approximation*: at each time t, the total state $\boldsymbol{\varrho}(t)$ is a factorized state with the "same" reduced state for the environment, explicitly

$$\forall t \in \mathbb{R}, \quad \tilde{\boldsymbol{\varrho}}(t) = \tilde{\rho}(t) \otimes \nu_t, \tag{I.32}$$

where $\nu_t = |0\rangle\langle 0|_t$ is the vacuum state of the bosonic operator B_t. Coupling with the corresponding vacuum state (at time t) implies that the average of the input modes canonical anticommutator is

$$\left\langle \left\{ b_t, b_{t'}^\dagger \right\} \right\rangle = \delta(t - t'), \tag{I.33}$$

where by definition the anticommutator of A and B is $\{A, B\} = AB + BA$. This equation and Eq. (I.25) form the Markovian assumptions of this framework [5]: the bosonic mode B_t is completely uncorrelated with the other modes at different times due to the delta-correlation of b_t. The global state evolves over an infinitesimal time $\mathrm{d}t$ as

$$\tilde{\boldsymbol{\varrho}}(t + \mathrm{d}t) = U_I(t + \mathrm{d}t, t) \overbrace{[\tilde{\rho}(t) \otimes \nu_t]}^{\tilde{\boldsymbol{\varrho}}(t)} U_I^\dagger(t + \mathrm{d}t, t), \tag{I.34}$$

where $U_I(t + \mathrm{d}t, t) \approx e^{-\frac{i}{\hbar}\tilde{H}_I(t)dt}$, as explained in Appendix B.

I.3.2 The Lindblad master equation

We just calculated the evolution of the state $\boldsymbol{\varrho}$ of the total system over an infinitesimal time $\mathrm{d}t$. In order to obtain the evolved state $\rho(t + \mathrm{d}t)$ of the system S, we want to "trace out" the environment (as explained in Section I.1.5) by applying the following partial trace

$$\tilde{\rho}(t + dt) = \mathrm{Tr}_E[\tilde{\boldsymbol{\varrho}}(t + \mathrm{d}t)]. \tag{I.35}$$

To write this expression into explicit terms, we develop the evolved total state and thus, we need to explicit the expression of $U_I(t + \mathrm{d}t, t)$. To do so, we can rewrite it using the environmental ladder operators, using Eq. (I.23), as

$$U_I(t + \mathrm{d}t, t) = \exp\left[\sqrt{k\mathrm{d}t}\Big(c \otimes B_t^\dagger - c^\dagger \otimes B_t\Big)\right]. \tag{I.36}$$

The subtlety here is that the unitary evolution operator scales with $\sqrt{\mathrm{d}t}$, and because $\mathrm{d}t = o(\sqrt{dt})$ expanding to order $\mathrm{d}t$ is already a fine description of the dynamics. Consequently, we expand the exponential to second order so that we capture all terms up to $\mathrm{d}t$, giving

$$\begin{aligned} U_I(t + \mathrm{d}t, t) \approx \mathbb{I} \otimes \mathbb{I} + \Big(c \otimes B_t^\dagger - c^\dagger \otimes B_t\Big)\sqrt{k\mathrm{d}t} + \\ \Big(c^{\dagger 2} \otimes B_t^2 + c^2 \otimes B_t^{\dagger 2} - cc^\dagger \otimes B_t^\dagger B_t - c^\dagger c \otimes B_t B_t^\dagger\Big)\frac{k\mathrm{d}t}{2}. \end{aligned} \tag{I.37}$$

We can now rewrite the evolution of $\tilde{\boldsymbol{\varrho}}$ during the infinitesimal time $\mathrm{d}t$, i.e. Eq. (I.34), using the previous expansion:

$$\begin{aligned} \tilde{\boldsymbol{\varrho}}(t + \mathrm{d}t) &\approx [\tilde{\rho}(t) \otimes |0\rangle\langle 0|_t] \times 1 + \left[\tilde{\rho}(t)c^\dagger \otimes |0\rangle\langle 1|_t + \mathrm{h.c.}\right] \times \sqrt{k\mathrm{d}t} \\ &+ \left[\frac{\sqrt{2}}{2}\tilde{\rho}(t)c^{\dagger 2} \otimes |0\rangle\langle 2|_t + \mathrm{h.c.} - \frac{1}{2}\tilde{\rho}(t)cc^\dagger \otimes |0\rangle\langle 0|_t + \mathrm{h.c.} + c\tilde{\rho}(t)c^\dagger \otimes |1\rangle\langle 1|_t\right] \times k\mathrm{d}t, \end{aligned} \tag{I.38}$$

where "h.c." stands for hermitian conjugate of the last term. We can now apply the partial trace Tr_E on the last equation. The evolved state of the system S is thus expressed as

$$\tilde{\rho}(t + \mathrm{d}t) = \tilde{\rho}(t) + k\left[c\tilde{\rho}(t)c^\dagger - \frac{c^\dagger c\tilde{\rho}(t) + \tilde{\rho}(t)c^\dagger c}{2}\right]\mathrm{d}t.$$

If we now form the difference quotient $\frac{\tilde{\rho}(t+\mathrm{d}t)-\tilde{\rho}(t)}{dt}$, in the infinitesimal limit, we find the differential equation followed by the state ρ at all time, i.e

$$\frac{\mathrm{d}\tilde{\rho}}{\mathrm{d}t}(t) = k\mathfrak{D}[c]\tilde{\rho}(t), \tag{I.39}$$

where we defined the super-operator $\mathfrak{D}[\hat{O}]\bullet \equiv \hat{O}\bullet\hat{O}^\dagger - \frac{\{\hat{O}^\dagger\hat{O},\bullet\}}{2}$ called the dissipator. Finally, by coming back into the Schrödinger representation, we find the so-called Lindblad master equation:

$$\boxed{\frac{\mathrm{d}\rho}{\mathrm{d}t}(t) = -\frac{i}{\hbar}[H_S, \rho(t)] + k\mathfrak{D}[c]\rho(t)}. \tag{I.40}$$

The proof is straightforward, however one should pay attention to the compensation of complex exponential when calculating $U_0cU_0^\dagger$, using the ansatz $\tilde{c}(t) = ce^{-i\omega_0(t-t_0)}$. Note that since ρ and c are operators on $\mathcal{H}_S$, any commutator with H_0 simplifies into a commutator with H_S.

Chapter II

Continuously measured open quantum systems

In the previous chapter, we went through the steps leading to the Lindblad master equation using the input-output formalism. In this derivation, the environment has been traced out to recover information on the system S from the global coupled system, leaving an impact on the dynamics of S, crystallized in the expression of a dissipator $\mathfrak{D}$. In the following we will focus on the impact of continuously monitoring the environment on the dynamics of the system S. Indeed, in the case of continuous measurement of the environment, the system state cannot follow the Lindblad master equation, we expect to find a differential equation that takes into account the impact of the measurement. However, we will see in the following that any differential equation obtained from continuously monitoring the environment is strongly linked to the Lindblad master equation.

In Chapter I, we have developed the input-output formalism in which the system S interacts sequentially in time with harmonic oscillators. Indeed, at each time the environment is represented by an harmonic oscillator represented by the annihilation operator B_t. This operator combines all environmental frequency modes by construction, as defined in Eq. (I.24). A fundamental property we have explained is that these operators are mutually uncorrelated. Consequently, the environment is "refreshed" after each evolution, ensuring no memory effects or temporal dependence of environmental operators. Therefore, initially the system S couples to B_t at time t, then the total system evolves according to the Lindblad master equation over the period of time $\mathrm{d}t$. Immediately after this evolution, at time $t + \mathrm{d}t$, the system uncouples from B_t to couple with a fresh $B_{t+\mathrm{d}t}$, evolving again, and so forth. In the following, we will go one step further by measuring the environment after each coupled evolution step, as schematized in Figure 1.

In this chapter, two kind of measurements will be presented: the photo-detection and the homodyne detection. After presenting the two measurement processes, we will tackle the notion of linear quantum trajectories. Finally, we will present a section of numerical simulations to illustrate the different measurement schemes.

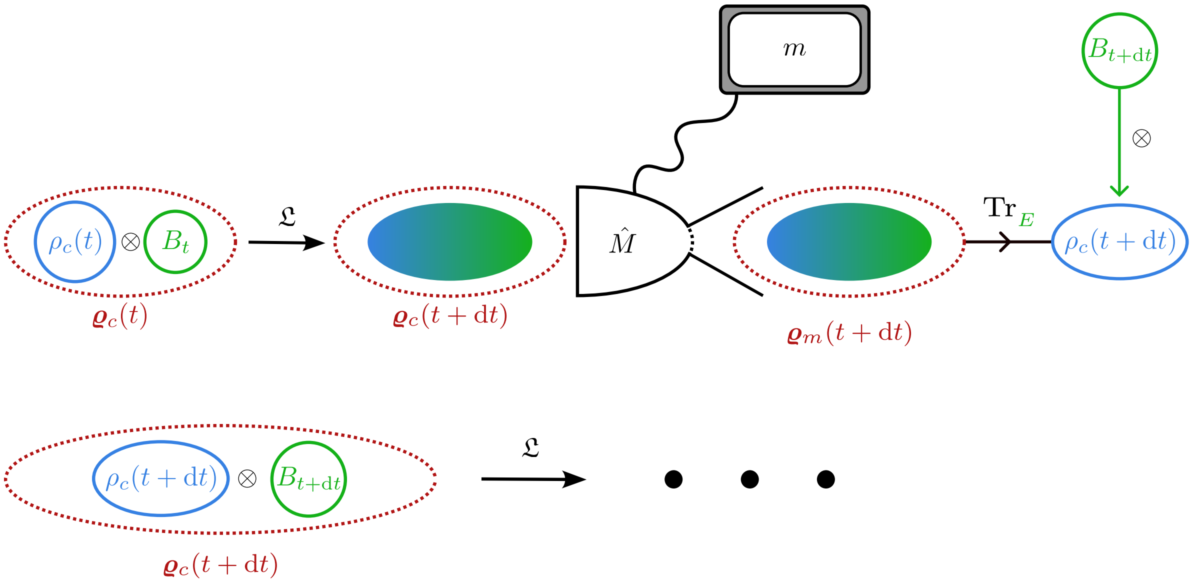

Figure 1: Scheme of continuous monitoring. In the first place the total system evolves from time t to $t + \mathrm{d}t$ according to the Lindblad master equation (denoted as $\mathfrak{L}$). We then measure the observable $\hat{M}$ on the total state and we denote the outcome as m. Then, by tracing out the environment, we find $\rho_c(t + \mathrm{d}t)$. To form the next total system $\varrho_c(t + \mathrm{d}t)$, the system couples instantaneously with the next temporal mode $B_{t+\mathrm{d}t}$, and so on. The process is repeated until the end of the monitoring. Note that before applying the partial trace, the system state is entangled with the environmental state due to the dynamics, this entanglement is represented by a gradient of color on the scheme. Moreover, it is important to realize that the index c at $t + \mathrm{d}t$ has been updated and includes the series of measurements up to time $t + \mathrm{d}t$ (and thus m obtained at time t in particular).

II.1 State of the art

II.2 Unconditional dynamics

The evolution of unmonitored open quantum systems is described by the Lindblad master equation, as seen in Section I.3. However, performing a measurement after each infinitesimal step makes the evolution dependent on each outcome, thereby altering the system's trajectory. If we consider the state of the system at time t, the state is conditional because of all the measurement outcomes from t_0 to t and is therefore noted as $\rho_c(t)$. The sequence of conditional states followed by a system is therefore called *a quantum trajectory* because the set of measurement outcomes gives a specific dynamical trajectory.

It is possible to rewrite the Lindblad master equation to yield the conditional states inside this unconditional equation [14]. This rewriting shows that the Lindblad master equation is decomposable into quantum trajectories that evolve in the absence of monitoring. Therefore, we refer to quantum trajectories as *an unravelling of the source dynamics* because it is one trajectory among the many paths coming from the same equation. In this sense, when describing photo-detection or homodyne detection, we unravel the master equation of a specific measurement. Nevertheless, **on average, they recover the same Lindblad master equation, i.e. the unconditional dynamics**. We will show this result in the case of photo-detection and homodyne detection below.

Note that in order to lighten notations, we stop specifying the interaction picture by $\tilde{\bullet}$ in the following. However, when a change of representation occurs it will be specified. Furthermore, we work from now on in natural units where $\hbar = 1$.

II.3 Photo-detection

In the case where the environment is continuously monitored by a photo-detector, the mathematical impact is a projection of the environment on the corresponding Fock basis $\{|n\rangle\langle n|_t\}_{n\in\mathbb{N}}$. Indeed, photo-detection is a way to measure the number operator $B_t^\dagger B_t$ because the Fock basis is the eigenbasis of this operator. At first glance the measurement would only give information about the environnement. However, by performing a partial trace Tr_E over the total evolved state, we can extract information on the system. Without the evolution of the couple system-environment, the partial trace on the evolved total state would not give any information on the system whatsoever.

In the following, we find the expression of the conditional state obtained at time $t+\mathrm{d}t$ in the case of photo-detection. Let $\{|m\rangle\langle m|_t\}_m$ be a PVM (see Section I.2), so that the measurement projects the environment at time t on the PVM element $|m\rangle\langle m|_t$. According to Eq. (I.34), the total evolved system is expressed as

$$\boldsymbol{\varrho}_c(t+\mathrm{d}t) = U_I(t+\mathrm{d}t,t)\overbrace{\left[\rho_c(t)\otimes|0\rangle\langle 0|_t\right]}^{\boldsymbol{\varrho}_c(t)}U_I^\dagger(t+\mathrm{d}t,t), \tag{II.1}$$

where the subscript c indicates that the system state is conditional. If the condition is obvious, we choose not to write this subscript explicitly, instead we write for example ρ_m if the outcome is m. Note that our convention of notation about conditional states means that at time $t+\mathrm{d}t$, we denote the state as $\rho_c(t)$ instead of $\rho_m(t)$; the symbol c encodes the list of all measurement outcomes. We observe that the state in Eq. (II.1) is indeed conditional because we consider the measurement already present in the evolution before time t. After the projection on $|m\rangle\langle m|_t$ and after applying the partial trace Tr_E, we find

$$\rho_m(t+\mathrm{d}t) = \frac{\langle m|_t\,\boldsymbol{\varrho}_c(t+\mathrm{d}t)\,|m\rangle_t}{p_j(t+\mathrm{d}t)}, \tag{II.2}$$

where $p_m(t+\mathrm{d}t) = \mathrm{Tr}\left[\boldsymbol{\varrho}_c(t+\mathrm{d}t)(\mathbb{I}\otimes|m\rangle\langle m|_t)\right]$. Eq. (I.38) shows that only projections on $|0\rangle\langle 0|_t$ and $|1\rangle\langle 1|_t$ are relevant to compute $\rho_m(t+\mathrm{d}t)$, as shown in the following.

Measurement result 0

From the expansion in Eq. (I.38), we find

$$\rho_0(t+\mathrm{d}t) = \frac{\langle 0|_t\,\boldsymbol{\varrho}_c(t+\mathrm{d}t)\,|0\rangle_t}{p_0(t+\mathrm{d}t)} \approx \frac{\rho_c(t) - \left\{\rho_c(t), c^\dagger c\right\}\frac{k\mathrm{d}t}{2}}{p_0(t+\mathrm{d}t)}. \tag{II.3}$$

The normalization is obtained using the same expansion, it gives

$$p_0(t+\mathrm{d}t) \approx 1 - \left\langle c^\dagger c\right\rangle_t k\mathrm{d}t, \tag{II.4}$$

where we have defined the average labelled by t as $\left\langle \hat{O}\right\rangle_t = \mathrm{Tr}\left[\rho_c(t)\hat{O}\right]$. In the infinitesimal limit, we expand the denominator in Eq. (II.2), obtaining

$$\rho_0(t+\mathrm{d}t) \approx \bar{\rho}_0(t+\mathrm{d}t)\times\left(1 - \left\langle c^\dagger c\right\rangle_t k\mathrm{d}t\right), \tag{II.5}$$

where we used the same notation as in Section I.2, $\bar{\rho}$ indicating an unnormalized state (here in the interaction picture). In particular, $\bar{\rho}_0(t+\mathrm{d}t)$ is the numerator of $\rho_0(t+\mathrm{d}t)$. Finally, by introducing the measurement super-operator $\mathfrak{H}[\hat{O}]\bullet = \hat{O}\bullet + \bullet\hat{O}^\dagger - \left\langle \hat{O} + \hat{O}^\dagger \right\rangle_t \bullet$, we can write at first order in $\mathrm{d}t$

$$\rho_0(t+\mathrm{d}t) = \rho_c(t) - \frac{k\mathrm{d}t}{2}\mathfrak{H}\left[c^\dagger c\right]\rho_c(t). \tag{II.6}$$

Measurement result 1

For this outcome, the same calculation method than the one used before still applies, and the conditional (unnormalized) state is expressed as

$$\bar{\rho}_1(t) = \langle 1|_t\, \boldsymbol{\varrho}(t+dt)\,|1\rangle_t \approx c\rho_c(t)c^\dagger k\mathrm{d}t. \tag{II.7}$$

The normalization is thus

$$p_1(t+\mathrm{d}t) = \left\langle c^\dagger c\right\rangle_t k\mathrm{d}t. \tag{II.8}$$

Combined, the two above equations give the normalized conditional state

$$\rho_1(t+\mathrm{d}t) = \frac{c\rho_c(t)c^\dagger}{\langle c^\dagger c\rangle_t}. \tag{II.9}$$

Mixing the results

From a statistical perspective, the measurement outcome at time $t+\mathrm{d}t$ is a Bernoulli-like process, yielding either 0 with probability $p_0(t+\mathrm{d}t)$ or 1 with probability $p_1(t+\mathrm{d}t)$. The full set of results up to time T is captured by the "sum" of results obtained between times t_0 and T. If for example we discretize time allowing to measure the system 30 times and that the sum of results is 20, it is obvious that it is composed of 10×0 outcomes and 20×1 outcomes. In this sense, we define the random variable $N(T,t_0) = \int_{t_0}^T \mathrm{d}N_t$, where in an infinitesimal slice of time $\mathrm{d}t$, the increment $\mathrm{d}N_t = N(t+\mathrm{d}t,t_0) - N(t,t_0)$ is either 0 or 1. This last observation leads to the fundamental relation: $\mathrm{d}N_t^2 = \mathrm{d}N_t$. In effect, $N(T,t_0)$ is a Poisson process and $\mathrm{d}N_t$ is a Poisson increment[7]. With the probabilities found above, we can calculate the stochastic average of $\mathrm{d}N_t$ to get

$$\mathbb{E}[\mathrm{d}N_t] = 0\times p_0(t+\mathrm{d}t) + 1\times p_1(t+\mathrm{d}t) = \left\langle c^\dagger c\right\rangle_t k\mathrm{d}t. \tag{II.10}$$

Note that the expression of $\mathbb{E}[\mathrm{d}N_t]$ is proportional to the expectation value $\langle c^\dagger c\rangle_t$. If we consider a system where c is the annihilation operator, this proportionality means that we can experimentally access to the mean occupation number. Therefore, the Poisson increment allows to write the differential conditional state $\mathrm{d}\rho_c(t+\mathrm{d}t)$ with respect to each possible outcome. In particular, $\mathrm{d}N_t = 0$ means that the differential conditional state is $\mathrm{d}\rho_0(t) = \rho_0(t+\mathrm{d}t) - \rho_c(t)$, while $\mathrm{d}N_t = 1$ means that $\mathrm{d}\rho_1(t) = \rho_1(t+\mathrm{d}t) - \rho_c(t)$. Consequently,

$$\mathrm{d}\rho_c(t) = \mathrm{d}N_t[\rho_1(t+\mathrm{d}t) - \rho_c(t)] + (1-\mathrm{d}N_t)[\rho_0(t+\mathrm{d}t) - \rho_c(t)]. \tag{II.11}$$

We can inject the results of Eqs. (II.6,II.9) in the last equation and by going back in the Schrödinger picture, we obtain

$$\boxed{\mathrm{d}\rho_c = -i[H_S,\rho_c]\mathrm{d}t - \frac{k}{2}\mathfrak{H}\left[c^\dagger c\right]\rho_c\mathrm{d}t + \left(\frac{c\rho_c c^\dagger}{\langle c^\dagger c\rangle_t} - \rho_c\right)\mathrm{d}N_t}. \tag{II.12}$$

[7] A Poisson increment follows a Poisson distribution with rate $\Lambda_t \in \mathbb{R}$: $\forall n \in \mathbb{N}$, $\mathbb{P}[\mathrm{d}N_t = n] = \frac{\Lambda_t^n}{n!}e^{-\Lambda_t}$. However, since we only have two possible outcomes (0 or 1), the rate is forced to be $\Lambda_t = 0$ because the sum of probabilities gives 1. By setting $\Lambda_t = \lambda_t\mathrm{d}t \to 0$ in the infinitesimal limit, we can find that $\mathbb{P}[\mathrm{d}N_t = 0] \approx 1 - \lambda_t\mathrm{d}t \to 1$ and $\mathbb{P}[\mathrm{d}N_t = 1] \approx \lambda_t\mathrm{d}t \to 0$. It is therefore trivial to show that $\mathbb{E}[\mathrm{d}N_t] = \lambda_t\mathrm{d}t$, and also that $\mathbb{V}[\mathrm{d}N_t] = \lambda_t\mathrm{d}t + \lambda_t^2\mathrm{d}t^2$.

This equation is called the photo-detection SME, note that we went into the Schrödinger picture to obtain this expression. Moreover, we have discarded terms of order $\mathrm{d}N_t\mathrm{d}t$ when calculating the expression of this SME. Indeed, we have shown that $\mathbb{E}[\mathrm{d}N_t] \propto \mathrm{d}t$, so $\mathbb{E}[\mathrm{d}N_t\mathrm{d}t] = (\mathrm{d}t)^2$, also since we can show that $\mathbb{V}[\mathrm{d}N_t] \propto \mathrm{d}t + o(\mathrm{d}t)$, it implies that $\mathbb{V}[\mathrm{d}N_t\mathrm{d}t] \propto (\mathrm{d}t)^2[\mathrm{d}t + o(\mathrm{d}t)] \propto \mathrm{d}t^3 + o(\mathrm{d}t^3)$. Thus, statistical terms as $\mathrm{d}N_t\mathrm{d}t$ are negligible compared to $\mathrm{d}t$ in average, Appendix C details why we can also neglect these terms locally. Finally, when derived for a pure state the expression simplifies to the Stochastic Schrödinger Equation (SSE)

$$\mathrm{d}\left|\psi_c\right\rangle = \left[-iH_S\mathrm{d}t + \frac{k}{2}\left(\left\langle c^\dagger c\right\rangle_t - c^\dagger c\right)\mathrm{d}t + \left(\frac{c}{\sqrt{\langle c^\dagger c\rangle_t}} - 1\right)\mathrm{d}N_t\right]\left|\psi_c\right\rangle, \tag{II.13}$$

which conserves the purity of the state during the stochastic evolution. Note that we can derive the SME from the SSE using Itō calculus rule [5]:

$$\mathrm{d}\rho_c = \mathrm{d}\left|\psi_c\right\rangle\left\langle\psi_c\right| + \left|\psi_c\right\rangle\mathrm{d}\left\langle\psi_c\right| + \mathrm{d}\left|\psi_c\right\rangle\mathrm{d}\left\langle\psi_c\right|. \tag{II.14}$$

The importance of the SME and of the SSE is double because these equations also contain in average the **unconditional** dynamics, as shown in the following.

We will now show briefly that we can write $\mathbb{E}[\rho_c] = \rho$, where ρ is the unconditional state following the Lindblad master equation. If we average the photo-detection SME over all possible trajectories we find

$$\begin{aligned}\mathbb{E}[\mathrm{d}\rho_c](t) = -i[H_S, R(t)]\mathrm{d}t - \frac{k}{2}\Big\{c^\dagger cR(t) + R(t)c^\dagger c - 2\mathbb{E}\Big[\left\langle c^\dagger c\right\rangle_t\rho_c(t)\Big]\Big\}\mathrm{d}t \\ + \mathbb{E}\left\{\left[\frac{c\rho_c(t)c^\dagger}{\langle c^\dagger c\rangle_t} - \rho_c(t)\right]\mathrm{d}N_t\right\},\end{aligned} \tag{II.15}$$

where $R(t) = \mathbb{E}[\rho_c(t)]$. To explicit the last term, we need the following relation $\mathbb{E}[f(\rho_c)(t)\mathrm{d}N_t] = k\mathbb{E}\big[\langle c^\dagger c\rangle_t f(\rho_c)(t)\big]\mathrm{d}t$, where f is a generic function [15]. So we obtain for the last term

$$\begin{aligned}\mathbb{E}\left\{\left[\frac{c\rho_c(t)c^\dagger}{\langle c^\dagger c\rangle_t} - \rho_c(t)\right]\mathrm{d}N_t\right\} &= k\mathbb{E}\left\{\left\langle c^\dagger c\right\rangle_t\left[\frac{c\rho_c(t)c^\dagger}{\langle c^\dagger c\rangle_t} - \rho_c(t)\right]\right\}\mathrm{d}t \\ &= kcR(t)c^\dagger\mathrm{d}t - k\mathbb{E}\Big[\left\langle c^\dagger c\right\rangle_t\rho_c(t)\Big]\mathrm{d}t.\end{aligned} \tag{II.16}$$

By injecting this result in Eq. (II.15), we find

$$\mathbb{E}[\mathrm{d}\rho_c](t) = -i[H_S, R(t)]\mathrm{d}t + k\mathfrak{D}[c]R(t)\mathrm{d}t, \tag{II.17}$$

which is exactly the Lindblad master equation for the state $R(t)$. We know that this equation is the expression of an unconditional state evolution, thus we can state that $\mathbb{E}[\rho_c(t)] = \rho(t)$. Therefore for photo-detection, we have shown that averaging over all the quantum trajectories reforms the unconditional Lindblad master equation, and that the average over all possibilities of the conditional state is the unconditional state. This property is also valid for homodyne detection [7, 16, 17] and will be used without proper demonstration in the following section.

II.4 Homodyne detection

In the previous section, we derived the SME in the case of photo-detection and showed its link to the Lindblad master equation. While photo-detection monitors the number of excitations, the homodyne detection targets field quadratures. This process involves mixing the signal with a strong local oscillator before measurement [7]. By measuring the global output signal, we

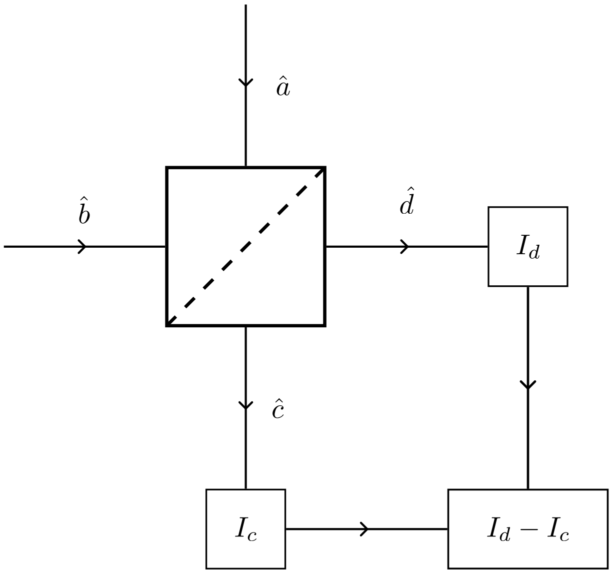

Figure 2: Schematic of the balanced homodyne detection method. The input $\hat{a}$ comes from the total system $S + E$ while $\hat{b}$ is a strong coherent field. The boxes represent photo-detectors measuring the fields after they passed through a beam-splitter (the box with a diagonal at the center). The apparatus calculates the difference of the photo-currents to obtain a (continuous) signal proportional to the quadrature of the system's field. We detail briefly this relationship in Appendix D.

have access to the quadrature of the physical system of interest [18]. We depict the homodyne detection scheme in Figure 2.

This kind of measurement corresponds to projecting the environmental oscillator — associated with B_t — on the eigenvectors of the quadrature operator $\hat{x}_\theta$. This operator is defined by $\hat{x}_\theta = (e^{-i\theta} B_t + e^{i\theta} B_t^\dagger)/\sqrt{2}$, where $\theta \in \mathbb{R}$, and verifies $\left[\hat{x}_\theta, \hat{x}_{\theta+\frac{\pi}{2}}\right] = i$. Even if it is not possible to construct normalized eigenstates of this operator, we can build the states $|x_\theta\rangle$ satisfying $\hat{x}_\theta |x_\theta\rangle = x_\theta |x_\theta\rangle$ in such a way that $\langle x_\theta | x_\lambda \rangle = \delta(\theta - \lambda)$, and verifying the closure relation $\int_{\mathbb{R}} |x_\theta\rangle\langle x_\theta| \, \mathrm{d}x_\theta = 1$. We will use the following overlapping property [19] between these states and the Fock states,

$$\langle x_\theta | n \rangle_t = \frac{e^{-\frac{x_\theta}{2}} H_n(x) e^{-in\theta}}{\sqrt{2^n n!} \pi^{\frac{1}{4}}}, \tag{II.18}$$

where $H_n(x)$ is the (physical) Hermite polynomial of order n. In particular, we will later use

$$\langle x_\theta | 0 \rangle_t = \pi^{-\frac{1}{4}} e^{-\frac{x_\theta^2}{2}}, \qquad \langle x_\theta | 1 \rangle_t = \sqrt{2} x_\theta e^{-i\theta} \langle x_\theta | 0 \rangle_t . \tag{II.19}$$

Note that since the state of the environnement at time t is the corresponding vacuum state, the probability to find the outcome x_θ at time t (before evolving) is

$$p_{x_\theta}(t) = |\langle x_\theta | 0 \rangle_t|^2 = \frac{e^{-x_\theta^2}}{\sqrt{\pi}}. \tag{II.20}$$

Continuous homodyne measurement corresponds by definition to the projection of the environment at each time on the quadrature states, giving the conditional (unnormalized) state

$$\begin{aligned} \bar{\rho}_c(t + \mathrm{d}t) &= \mathrm{Tr}_E \left[U_I(t + \mathrm{d}t, t) \boldsymbol{\varrho}_c(t) U_I^\dagger(t + \mathrm{d}t, t) (\mathbb{I} \otimes |x_\theta\rangle\langle x_\theta|) \right] \\ &= \langle x_\theta | U_I(t + \mathrm{d}t, t) [\rho_c(t) \otimes |0\rangle\langle 0|_t] U_I^\dagger(t + \mathrm{d}t, t) | x_\theta \rangle , \end{aligned} \tag{II.21}$$

with the corresponding normalization (probability)

$$p_{x_\theta}(t+\mathrm{d}t) = \mathrm{Tr}_S[\bar{\rho}_c(t+\mathrm{d}t)]. \tag{II.22}$$

We will now use the expansion of Eq. (I.38) to develop Eq. (II.21). However, we truncate it to $o\Big(\sqrt{dt}\Big)$ because $\mathrm{d}t$ is negligible compared to $\sqrt{\mathrm{d}t}$, giving for the conditional unnormalized evolved state

$$\bar{\rho}_c(t+\mathrm{d}t) = |\langle x_\theta|0\rangle_t|^2\rho_c(t) + \Big[c\rho_c(t)\,\langle x_\theta|1\rangle_t\,\langle 0|x_\theta\rangle + \rho_c(t)c^\dagger\,\langle x_\theta|0\rangle_t\,\langle 1|x_\theta\rangle\Big]\sqrt{k\mathrm{d}t} + o\Big(\sqrt{\mathrm{d}t}\Big). \tag{II.23}$$

Simplifying this expression and factorizing it, using Eqs. (II.19,II.20), we find

$$\bar{\rho}_c(t+\mathrm{d}t) = p_{x_\theta}(t)\Big\{\rho_c(t) + x_\theta\sqrt{2k\mathrm{d}t}\Big[c\rho_c(t)e^{-i\theta} + \rho_c(t)c^\dagger e^{i\theta}\Big]\Big\} + o\Big(\sqrt{dt}\Big), \tag{II.24}$$

with the normalization

$$p_{x_\theta}(t+\mathrm{d}t) = p_{x_\theta}(t)\Big\{1 + x_\theta\sqrt{2k\mathrm{d}t}\,\Big\langle ce^{-i\theta} + c^\dagger e^{i\theta}\Big\rangle_t\Big\} + o\Big(\sqrt{\mathrm{d}t}\Big) \tag{II.25}$$

$$\approx \frac{1}{\sqrt{\pi}} \times \exp\left\{-\left[x_\theta - \sqrt{\frac{k\mathrm{d}t}{2}}\,\Big\langle ce^{-i\theta} + c^\dagger e^{i\theta}\Big\rangle_t\right]^2\right\}. \tag{II.26}$$

We can deduce from Eq. (II.26) that the probability to find the result x_θ at time $t+\mathrm{d}t$ corresponds to a Gaussian distribution[8] (up to order $\sqrt{\mathrm{d}t}$). In this sense, the outcome of measurement is a random variable such that $x_\theta \hookrightarrow \mathcal{N}\left(\sqrt{\frac{k\mathrm{d}t}{2}}\,\langle ce^{-i\theta} + c^\dagger e^{i\theta}\rangle_t, \frac{1}{2}\right)$. By defining the random variable $\mathrm{d}y_t = \sqrt{2\mathrm{d}t}x_\theta$, we find $\mathbb{E}[\mathrm{d}y_t] = \sqrt{k}\,\langle ce^{-i\theta} + c^\dagger e^{i\theta}\rangle_t\,\mathrm{d}t$ and $\mathbb{V}[\mathrm{d}y_t] = \mathrm{d}t$. Consequently, we can statistically express this random variable as

$$\mathrm{d}y_t = \sqrt{k}\,\Big\langle ce^{-i\theta} + c^\dagger e^{i\theta}\Big\rangle_t\,\mathrm{d}t + \mathrm{d}w_t, \tag{II.27}$$

where $\mathrm{d}w_t$ follows a Gaussian distribution characterized by $\mathbb{E}[\mathrm{d}w_t] = 0$ and $\mathbb{V}[\mathrm{d}w_t] = \mathrm{d}t$. Such a random variable is called a classical Wiener increment and obeys the Itō rule $\mathrm{d}w_t^2 = dt$. This rule should be understood in the distributional sense as $\int_0^t \mathrm{d}w_{t'}^2 = \int_0^t \mathrm{d}t'$ (valid for any interval $[0,t]$) [16]. Experimentally, we obtain at time t a photocurrent $I(t)$ from the apparatus of homodyne detection process. To make this photocurrent appear, we define the integrated result of homodyne detection at time T as

$$y_T = \int_0^T \mathrm{d}y_t = \int_0^T I(t)\mathrm{d}t. \tag{II.28}$$

Using Eq. (II.27), we can define the photocurrent in the distributional sense as

$$I(t) = \sqrt{k}\,\Big\langle ce^{-i\theta} + c^\dagger e^{i\theta}\Big\rangle_t + \xi(t), \tag{II.29}$$

where $\xi(t) = \frac{\mathrm{d}w_t}{\mathrm{d}t}$ is a classical white noise such that $\mathbb{E}[\xi(t)\xi(t')] = \delta(t-t')$ [5]. The definition of $\xi(t)$ implies $\mathbb{E}[\xi(t)] = 0$ and $\mathbb{V}[\xi(t)] = \frac{1}{\mathrm{d}t} \to +\infty$ (due to the Itō rule). However, when integrated inside of Eq. (II.28), the variance becomes a finite quantity. Moreover, the notation seems to define ξ as the derivative of w_t which is wrong because a Wiener process is not differentiable anywhere. Indeed, we can see it unformally by forming the rate of change $\frac{W_{t+\mathrm{d}t}-W_t}{\mathrm{d}t} = \frac{\mathrm{d}w_t}{\mathrm{d}t} \sim$

[8]A Gaussian or normal distribution of mean μ and variance σ^2 is characterized by the following probability density function: $X \mapsto \frac{1}{\sqrt{2\pi\sigma^2}}\exp\left[-\frac{(X-\mu)^2}{2\sigma^2}\right]$. We denote by $X \hookrightarrow \mathcal{N}(\mu,\sigma^2)$ the fact that the random variable X follows a normal distribution with mean μ and variance σ^2.

$\frac{1}{\sqrt{\mathrm{d}t}} \underset{\mathrm{d}t\to 0}{\longrightarrow} +\infty$. The correct usage of ξ is really as an integrand multiplied by $\mathrm{d}t$, as in Eq. (II.28). **Note that, up to a factor $1/\sqrt{2}$, the photocurrent $I(t)$ contains the system's average quadrature.** Indeed, the average of the quadrature operator for the system S is defined as

$$\left\langle \frac{ce^{-i\theta} + c^\dagger e^{i\theta}}{\sqrt{2}} \right\rangle_t . \tag{II.30}$$

To find the homodyne SME, we inject Eq. (II.27) into Eqs. (II.24,II.25). By dividing these new equations, we form the normalized evolved state. We can then expand the denominator and find

$$\mathrm{d}\rho_c(t) = \sqrt{k}\mathfrak{H}\left[ce^{-i\theta}\right]\rho_c(t)\mathrm{d}w_t + o(\sqrt{\mathrm{d}t}). \tag{II.31}$$

The terms of order $o(\mathrm{d}t)$ are inferred as we know that the average over all trajectories must recover the Lindblad master equation, as explained in the last paragraph of Section II.3. However, the average of Eq. (II.31) trivially gives 0, so the correct SME (in the Schrödinger picture) is

$$\boxed{\mathrm{d}\rho_c = -i[H_S, \rho_c]\mathrm{d}t + k\mathfrak{D}[c]\rho_c\mathrm{d}t + \sqrt{k}\mathfrak{H}\left[ce^{-i\theta}\right]\rho_c\mathrm{d}w_t}, \tag{II.32}$$

where we recall that $\mathrm{d}w_t = \mathrm{d}y_t - \sqrt{k}\left\langle ce^{-i\theta} + c^\dagger e^{i\theta}\right\rangle_t \mathrm{d}t$.

If we consider the previous case of measurement, the photo-detection, we observe that the unravelling obeys to a jump-like process. Indeed, the integral $N(T) = \int_0^T \mathrm{d}N_t$ consists in a sum of quantities jumping from 0 to 1. Instead, the homodyne unravelling is following a diffusive process because $W(T) = \int_0^T \mathrm{d}w_t$ is a Gaussian white noise with variance T, thus linearly increasing with time. In this sense, the same Lindblad master equation can be unravelled via two different physical process (jump-like and diffusive), resulting in two different SMEs (photo-detection and homodyne detection respectively).

II.5 Linear quantum trajectories

The previously obtained SMEs are non-linear in ρ_c, implying heavy computational work in order to simulate a trajectory. Indeed, if we observe the two SMEs obtained in the previous sections, we can remark their non-linearity with respect to $\rho_c(t)$. The photo-detection SME, i.e. Eq. (II.12), is non-linear because of the term $\left\langle c^\dagger c\right\rangle_t = \mathrm{Tr}\left[c^\dagger c\rho_c(t)\right]$ at the denominator in the term multiplying $\mathrm{d}N_t$, and also because it is multiplied by $\rho_c(t)$ in the expression of $\mathfrak{H}\left[c^\dagger c\right]\rho_c(t)$. Similarly, in Eq. (II.32), the homodyne SME is non-linear because the term $\left\langle ce^{-i\theta} + c^\dagger e^{i\theta}\right\rangle_t = \mathrm{Tr}\left[(ce^{-i\theta} + c^\dagger e^{i\theta})\rho_c(t)\right]$ is squared when we multiply $\mathfrak{H}[ce^{-i\theta}]\rho_c(t)$ by $\mathrm{d}w_t$. However, linear SMEs can be built at the cost of not using normalized states as we explain in the following.

To find the homodyne linear master equation, we replace $\left\langle ce^{-i\theta} + c^\dagger e^{i\theta}\right\rangle_t$ by $\mu \in \mathbb{R}$ in Eq. (II.32). In particular, the replaced term is also involved in the normalization of the state at time $t + \mathrm{d}t$, as shown in Eq. (II.26). Therefore, the linear equation describes the evolution of non-normalized states, denoted as $\breve{\rho}_c(t, \mu)$. We distinguish these non-normalized states from the ones introduced in Section I.2 noted $\bar{\rho}$ because their trace is not equal to the probability to find this trajectory, as it will be made clear further below. If we observe the effect of changing $\left\langle ce^{-i\theta} + c^\dagger e^{i\theta}\right\rangle_t$ into a constant, we realize that the homodyne outcome becomes $\mathrm{d}y_t = \sqrt{k}\mu\mathrm{d}t + \mathrm{d}w_t$. Therefore, a natural choice is to set $\mu = 0$, yielding $\mathrm{d}y_t = \mathrm{d}w_t$. Consequently, the linear SME is expressed as

$$\mathrm{d}\breve{\rho}_c = -i[H_S, \breve{\rho}_c]\mathrm{d}t + k\mathfrak{D}[c]\breve{\rho}_c\mathrm{d}t + \sqrt{k}\Big(ce^{-i\theta}\breve{\rho}_c + \breve{\rho}_c c^\dagger e^{i\theta}\Big)\mathrm{d}y_t, \tag{II.33}$$

where we chose to simplify the notation $\breve{\rho}_c(t, \mu = 0) = \breve{\rho}_c(t)$. This SME allows one to calculate iteratively $\breve{\rho}_c(t + \mathrm{d}t) = \breve{\rho}_c(t) + \mathrm{d}\breve{\rho}_c(t)$ for each timestep by simply generating a classical white noise $\mathrm{d}w_t$. Then, by dividing $\breve{\rho}_c(t+\mathrm{d}t)$ by its trace, one can find the expected (normalized) state $\rho_c(t + \mathrm{d}t)$.

In fact, building the linear SME introduces a different probability law than the one truly followed by the outcomes and states in the normalized case. Indeed, the outcome $\mathrm{d}y_t$ has to follow a different probability distribution when it is simply equal $\mathrm{d}w_t$ (in the case of the linear SME). We denote as p_{true} the distribution which gives the relationship $\mathbb{E}_{p_{\text{true}}}[\rho_c(t)] = \rho(t)$, and as p_{ost} the distribution which gives $\mathbb{E}_{p_{\text{ost}}}[\breve{\rho}_c(t)] = \rho(t)$ ("ost" stands for ostensible [15]). The relationship between these two probabilities is

$$p_{\text{true}}(\text{traj.}) = p_{\text{ost}}(\text{traj.}) \times \operatorname{Tr}\left[\breve{\rho}_c(t)\right], \tag{II.34}$$

where "traj." stands for the quantum trajectory whose state at time t is $\breve{\rho}_c(t)$. This last equation shows how the trace of the solution of the linear SME encodes the (true) probability of the conditional state $\rho_c(t)$. If we explicit the expectation values according to the distributions p_{true} and p_{ost}, one can show that Eq. (II.34) can be used to prove[9]that $\mathbb{E}_{p_{\text{true}}}[\rho_c(t)] = \mathbb{E}_{p_{\text{ost}}}[\breve{\rho}_c(t)]$. Moreover, we have shown in the previous sections that $\mathbb{E}_{p_{\text{true}}}[\rho_c(t)] = \rho(t)$, so $\rho(t) = \mathbb{E}_{p_{\text{true}}}[\rho_c(t)] = \mathbb{E}_{p_{\text{ost}}}[\breve{\rho}_c(t)]$. In this sense, we can follow the evolution of the unconditional state from the non-normalized state evolution.

Likewise, in the case of photo-detection, the results in Eq. (II.34) hold and we can repeat the same kind of linearization method used in the previous paragraph. Therefore, one can obtain

$$\mathbb{E}_{p_{\text{ost}}}[\mathrm{d}N_t] = \mu' k \mathrm{d}t, \tag{II.35}$$

by replacing $\langle c^\dagger c \rangle_t$ by μ' in the photo-detection SME, i.e. in Eq. (II.12). The natural choice seems to be $\mu' = 1$, so that the expectation value of $\mathrm{d}N_t$ (calculated with p_{ost}) is simply $k\mathrm{d}t$. It follows that the linear SME for photo-detection is

$$\mathrm{d}\breve{\rho}_c = -i[H_S, \breve{\rho}_c]\mathrm{d}t - \frac{k}{2}\left\{c^\dagger c, \breve{\rho}_c\right\}\mathrm{d}t + k\breve{\rho}_c\mathrm{d}t + \left(c\breve{\rho}_c c^\dagger - \breve{\rho}_c\right)\mathrm{d}N_t. \tag{II.36}$$

In the simulations we present in the last section of this chapter, we could use these linear quantum trajectories. However, we chose to use a Monte Carlo method that will be introduced in the following section, also, we will use a non-linear solver available in the `QuantumToolbox.jl` package.

II.6 Evolution in terms of Kraus operators

In the literature, one of the best solution to simulate the photo-detection process, which we developped in Section II.3, is the so-called Monte Carlo method [15]. Nevertheless, we need to use Kraus operators in order to find the expression needed in the Monte Carlo simulation. Let us tackle the photo-detection case at first. Indeed, the results of the QMT Section I.2 are going to be use hereafter.

II.6.1 Photo-detection

We define the Kraus operators yielding the result 0 as

$$K_0(\mathrm{d}t) = \mathbb{I} - \left(k\frac{c^\dagger c}{2} + iH_S\right)\mathrm{d}t, \tag{II.37}$$

[9]We can explicit the expression of the expected value of a continuous random variable, such that $X : \Omega \to \mathbb{R}$ with probability density function p, as $\mathbb{E}_p[X] = \int_\Omega X \mathrm{d}P = \int_{\mathbb{R}} x p(x)\mathrm{d}x$. In the case of the states (normalized or not), we should integrate over all possible trajectories (so every possible measurement outcome).

together with its adjoint $K_0^\dagger(\mathrm{d}t)$. For the result 1, we define

$$K_1(\mathrm{d}t) = \sqrt{k\mathrm{d}t}c, \tag{II.38}$$

and its adjoint $K_1^\dagger(\mathrm{d}t)$. We recall that the POVM must respect

$$K_0^\dagger(\mathrm{d}t)K_0(\mathrm{d}t) + K_1^\dagger(\mathrm{d}t)K_1(\mathrm{d}t) = \mathbb{I} + o(\mathrm{d}t). \tag{II.39}$$

Indeed, this last equation is verified at fist order in $\mathrm{d}t$ using the definition of Kraus operators. Thus, the ensemble of Kraus operators is a valid POVM. We can look at the probabilities to find each outcome with respect to the Kraus operators formalism. We can show that

$$p_0(\mathrm{d}t) = \mathrm{Tr}\left[K_0^\dagger(\mathrm{d}t)\rho_c(t)K_0(\mathrm{d}t)\right] = 1 - k\left\langle c^\dagger c\right\rangle_t \mathrm{d}t + o(\mathrm{d}t). \tag{II.40}$$

Also we calculate the probability of outcome 1, giving

$$p_1(\mathrm{d}t) = \mathrm{Tr}\left[K_1^\dagger(\mathrm{d}t)\rho_c(t)K_1(\mathrm{d}t)\right] = k\left\langle c^\dagger c\right\rangle_t \mathrm{d}t + o(\mathrm{d}t). \tag{II.41}$$

Therefore, we can calculate $p_0(\mathrm{d}t) + p_1(\mathrm{d}t) = 1 + o(\mathrm{d}t)$ as expected. One can observe from Eq. (II.10) that $p_1(\mathrm{d}t) = \mathbb{E}[\mathrm{d}N_t]$ in particular. We now rewrite the evolved conditional state with these Kraus operators and with the Poisson increment $\mathrm{d}N_t$ as

$$\rho_c(t+\mathrm{d}t) = \underbrace{\frac{K_0(\mathrm{d}t)\rho_c(t)K_0^\dagger(\mathrm{d}t)}{p_0(\mathrm{d}t)}}_{\rho_0(t+\mathrm{d}t)}(1-\mathrm{d}N_t) + \underbrace{\frac{K_1(\mathrm{d}t)\rho_c(t)K_1^\dagger(\mathrm{d}t)}{p_1(\mathrm{d}t)}}_{\rho_1(t+\mathrm{d}t)}\mathrm{d}N_t. \tag{II.42}$$

One can develop this expression up to $\mathrm{d}t$ in the same way than in Section II.3 to find the SME, i.e Eq. (II.12). We will ommit the $o(\mathrm{d}t)$ in the following to lighten notations.

This new formulation is particularly interesting because the unnormalized expressions of evolved states (depending on each outcome) are

$$\tilde{\rho}_0(t+\mathrm{d}t) = K_0(\mathrm{d}t)\tilde{\rho}_c(t)K_0^\dagger(\mathrm{d}t), \qquad \tilde{\rho}_1(t+\mathrm{d}t) = K_1(\mathrm{d}t)\tilde{\rho}_c(t)K_1^\dagger(\mathrm{d}t). \tag{II.43}$$

Explicitely, by using Eqs (II.37,II.38), we get

$$\tilde{\rho}_0(t+\mathrm{d}t) = \tilde{\rho}_c(t) - \frac{k}{2}\left\{c^\dagger c, \tilde{\rho}_c(t)\right\}\mathrm{d}t - i[H_S, \tilde{\rho}_c(t)]\mathrm{d}t, \qquad \tilde{\rho}_1(t+\mathrm{d}t) = kc\tilde{\rho}_c(t)c^\dagger\mathrm{d}t. \tag{II.44}$$

The evolved system is in the state $\tilde{\rho}_0(t+\mathrm{d}t)$ with the probability $p_0(\mathrm{d}t)$ or in the state $\tilde{\rho}_1(t+\mathrm{d}t)$ with the probability $p_1(\mathrm{d}t)$. Moreover, we recall that $p_0(\mathrm{d}t) \gg p_1(\mathrm{d}t)$ in the infinitesimal limit and so the system evolves almost everytime according to the outcome 0, we call this the "no-jump" evolution. In the following, we will be interested into simulating SSEs for simplicity and rapidity. Therefore, here are the conditional evolved states for each outcome

$$\left|\tilde{\psi}_0(t+\mathrm{d}t)\right\rangle = K_0(\mathrm{d}t)\left|\tilde{\psi}_c(t)\right\rangle = \left[\mathbb{I} - \left(k\frac{c^\dagger c}{2} + iH_S\right)\mathrm{d}t\right]\left|\tilde{\psi}_c(t)\right\rangle, \tag{II.45}$$

$$\left|\tilde{\psi}_1(t+\mathrm{d}t)\right\rangle = K_1(\mathrm{d}t)\left|\tilde{\psi}_c(t)\right\rangle = \sqrt{k\mathrm{d}t}c\left|\tilde{\psi}_c(t)\right\rangle. \tag{II.46}$$

The Monte Carlo method [15] consists in allowing the state to evolve using K_0 Kraus operator all the time, unless a jump occurs. Explicitely, the differential equation of no-jump evolution is

$$\frac{\mathrm{d}}{\mathrm{d}t}\left|\tilde{\psi}(t)\right\rangle = -\left(k\frac{c^\dagger c}{2} + iH_S\right)\left|\tilde{\psi}(t)\right\rangle. \tag{II.47}$$

To determine when a jump occurs, we first need to pick a random number R before the no-jump evolution. The effect of the non-normalized differential equation is notably to diminish the value of the normalization with respect to time. Indeed,

$$\mathrm{d}\left\langle \tilde{\psi}(t) \middle| \tilde{\psi}(t) \right\rangle = -k \left\langle c^\dagger c \right\rangle_t \mathrm{d}t, \tag{II.48}$$

where we used the notation $\left\langle c^\dagger c \right\rangle_t = \left\langle \tilde{\psi}(t) \middle| (c^\dagger c) \middle| \tilde{\psi}(t) \right\rangle$. Therefore, this last equation proves that the square of the state's norm is diminishing linearly with respect to time. The Monte Carlo process uses this fact to solve Eq. (II.47) for a time T such that $\left\langle \tilde{\psi}(T) \middle| \tilde{\psi}(T) \right\rangle = R$. The time T is the "jump" time at which we must apply the evolution using Kraus operator K_1. The state is then normalized and we act like at the beginning: picking a new random number R, the no-jump evolution and so on. Note that applying the normalized evolution with outcome 1 is the same as applying the unnormalized evolution and then normalizing the state.

II.6.2 Homodyne detection

Following the same concepts than in the photo-detection case, we can define the Kraus operators responsible for the evolution according to the homodyne measurement. In this sense, we define

$$K_{\mathrm{d}y_t} = \mathbb{I} - iH_S \mathrm{d}t - \frac{k}{2} c^\dagger c \mathrm{d}t + \sqrt{k} c \mathrm{d}y_t, \tag{II.49}$$

with its adjoint $K_{\mathrm{d}y_t}^\dagger$. If we calculate the sum of POVM elements, we find

$$\int K_J^\dagger K_J \mathrm{d}J \neq \mathbb{I} + o(\mathrm{d}t), \tag{II.50}$$

where we denoted $J = \mathrm{d}y_t$ for convenience. This integral should be understood by the sentence "we sum the POVM elements for every possible outcome $\mathrm{d}y_t$". Indeed, $K_J^\dagger K_J = \mathbb{I} + \sqrt{k}(ce^{-i\theta} + c^\dagger e^{i\theta})J + o(dt)$ is the reason why the normalization of POVM fails here. To obtain the correct normalization, one must calculate the integral according to the ostensible probability (see Section II.5), thus

$$\int K_J^\dagger K_J p_{\mathrm{ost}}(J) \mathrm{d}J = \mathbb{I} + o(\mathrm{d}t). \tag{II.51}$$

This interpretation of "Kraus operator" is more general than the one initially introduced in the QMT section in the sense that the normalization is not respected according to the true probability p_{true}. This reasoning about ostensible probability can be directly obtained by realizing that the unnormalized evolved state is

$$\tilde{\rho}_c(t + \mathrm{d}t) = K_{\mathrm{d}y_t} \rho_c(t) K_{\mathrm{d}y_t}^\dagger = \breve{\rho}_c(t + \mathrm{d}t), \tag{II.52}$$

where $\breve{\rho}_c(t + \mathrm{d}t)$ is the solution of the unnormalized linear SME. In particular, we can derive it from Eq. (II.33). Therefore, this justifies employing the ostensible probability when dealing with such states.

II.7 Simulations

Conclusion

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Appendix A

Superposition of states VS statistical mixture

In order to illustrate the concepts of superposition of states and statistical mixture, let us consider to observables A, B such that

$$\forall i \in \{1,2\}, \quad A|\phi_i\rangle = a_i|\phi_i\rangle,$$
$$\forall i \in \{1,2\}, \quad B|\psi_i\rangle = b_i|\psi_i\rangle.$$

We can define a state $|\Psi\rangle$ in superposition of the eigenstates of the observable B:

$$\boxed{|\Psi\rangle = \lambda_1|\psi_1\rangle + \lambda_2|\psi_2\rangle}, \tag{A. 1}$$

where $(\lambda_1, \lambda_2) \in \mathbb{C}^2$ such that $|\lambda_1|^2 + |\lambda_2|^2 = 1$. We thus observe the result b_i with following probability

$$\forall i \in \{1,2\}, \quad \mathbb{P}_\Psi[b_i] = |\langle\psi_i|\Psi\rangle|^2 = |\lambda_i|^2 \equiv p_i. \tag{A. 2}$$

If now we measure the system in order to find one of the eigenvalues of A, a_1 for example, we find

$$\begin{aligned}\mathbb{P}_\Psi[a_1] &= |\langle\phi_1|\Psi\rangle|^2 \\ &= |\lambda_1\langle\phi_1|\psi_1\rangle + \lambda_2\langle\phi_1|\psi_2\rangle|^2.\end{aligned} \tag{A. 3}$$

Moreover, by observing that $\forall i \in \{1,2\}, \quad \mathbb{P}_{\psi_i}[a_1] = |\langle\phi_1|\psi_i\rangle|^2$, we finally find that

$$\boxed{\mathbb{P}_\Psi[a_1] = p_1\mathbb{P}_{\psi_1}[a_1] + p_2\mathbb{P}_{\psi_2}[a_1] + 2\,\mathrm{Re}\{\lambda_1\lambda_2^*\langle\phi_1|\psi_1\rangle\langle\phi_1|\psi_2\rangle^*\}}. \tag{A. 4}$$

We will now prepare a state with the same probabilities $\mathbb{P}[b_i] \equiv p_i$ but which is completely different.

A statistical mixture can be well understood by the following thought experiment:

- We want to prepare N states in total using the eigenstates $\{\psi_1, \psi_2\}$ of B.
- We prepare n_1 states in $|\psi_1\rangle$.
- We prepare n_2 states in $|\psi_2\rangle$, so $n_1 + n_2 = N$.

At this point, we have a set of N systems that we can use. However, if we pick a state randomly the probability to pick $|\psi_i\rangle$ depends on the fraction of it $p_i = \frac{n_i}{N}$. This fraction p_i coincides theoretically with the previous probability $P_\Psi[b_i]$ for the superposition of states: it represents the probability to find the state $|\psi_i\rangle$ from the general state. If we now use the observable A to measure the system we picked, the probability to find the result a_1 becomes

$$\mathbb{P}_{\text{sm}}[a_1] = \frac{n_1\mathbb{P}_{\psi_1}[a_1] + n_2\mathbb{P}_{\psi_2}[a_1]}{N} = p_1\mathbb{P}_{\psi_1}[a_1] + p_2\mathbb{P}_{\psi_2}[a_1],$$

by using the law of total probability. Therefore,

$$\boxed{\mathbb{P}_{\text{sm}}[a_1] = p_1\mathbb{P}_{\psi_1}[a_1] + p_2\mathbb{P}_{\psi_2}[a_1]}. \tag{A. 5}$$

By comparing the last two boxed equations, one can realize that the mathematical expression for the same probability is written differently depending on if we use a superposition of states or a statistical mixture. However, the probability of the statistical mixture is present in the expression of probability of the superposition of states. This difference of formula for the same probability therefore proves that the Dirac formalism $|\psi\rangle$ cannot represent the kind of system whose probability comes from a statistical mixture. It is in fact impossible to represent a statistical mixture with a pure state. Instead, we need to use the density matrix

$$\rho_{\text{sm}} = p_1 |\psi_1\rangle\langle\psi_1| + p_2 |\psi_2\rangle\langle\psi_2| , \tag{A. 6}$$

so that,

$$\mathbb{P}_{\rho_{\text{sm}}}[a_1] = p_1\mathbb{P}_{\psi_1}[a_1] + p_2\mathbb{P}_{\psi_2}[a_1], \tag{A. 7}$$

because by definition $\mathbb{P}_{\rho_{\text{sm}}}[a_1] = \text{Tr}\left[\rho_{\text{sm}} |\phi_1\rangle\langle\phi_1|\right]$.

Appendix B

Representations of quantum mechanics

Brief complement of quantum mechanics

An average is defined in quantum mechanics by

$$\begin{aligned} \langle O \rangle_\psi &= \langle \psi | O | \psi \rangle \,, \\ \langle O \rangle_\rho &= \mathrm{Tr}\left[\rho O\right] , \end{aligned}$$

where the subscripts $|\psi\rangle$ and ρ refer respectively to the representation of pure states or to density matrix more general formalism.

B.1 Schrödinger and Heisenberg pictures

In the Schrödinger picture we saw that $|\psi(t)\rangle = U(t)\,|\psi\rangle$, where $U(t) = e^{-\frac{i}{\hbar}Ht}$ in the case of a time-independent Hamiltonian with $t_0 = 0$. Nevertheless, the following results can be derived in the general case of a time dependent Hamiltonian, [3]. Considering the average value of an observable $A_S(t)$ in the Schrödinger picture (in this picture, the time-dependence of an observable can only be explicit), we find

$$\begin{aligned} \langle A(t) \rangle &= \langle \psi(t) | A_S(t) | \psi(t) \rangle \\ &= \langle \psi(t_0) | U^\dagger(t,t_0) A_S(t) U(t,t_0) | \psi(t_0) \rangle \qquad \text{(B. 1)} \\ &= \langle \psi | A_H(t) | \psi \rangle_H \,. \qquad \text{(B. 2)} \end{aligned}$$

This calculation naturally reveals the Heisenberg representation of observables and states. Indeed, we have that $|\psi(t_0)\rangle_S \equiv |\psi\rangle_H$ which links both pictures at the initial time t_0. Furthermore, we can link both representation at all time by calculating the evolution of an observable in the Heisenberg picture:

$$\boxed{\frac{\mathrm{d}A_H}{\mathrm{d}t} = \frac{i}{\hbar}\left[H_H, A_H\right] + \frac{\partial A_S}{\partial t}}. \qquad \text{(B. 3)}$$

This equation leads to the famous *Ehrenfest theorem* analogous in quantum mechanics:

$$\boxed{\frac{\mathrm{d}\left\langle A \right\rangle}{\mathrm{d}t} = \frac{i}{\hbar}\left\langle [H(t), A(t)] \right\rangle + \left\langle \frac{\partial A}{\partial t} \right\rangle}. \qquad \text{(B. 4)}$$

We observe that if the observable doesn't depend explicitly on time, then the dynamics of the operator is fully described in the Heisenberg picture.

B.2 Interaction picture

As mentioned in the Section I.3, we can represent the coupling between S and E with an interaction Hamiltonian such that

$$H_T = \overbrace{H_S \otimes \mathbb{I}_E + \mathbb{I}_S \otimes H_E}^{H_0} + H_I. \tag{B. 1}$$

From now on, we denote the state of the total system as $\boldsymbol{\varrho}$. Knowing that $|\psi(t)\rangle = U(t,t_0)\,|\psi\rangle$, we can write that

$$\boldsymbol{\varrho}(t) = U(t,t_0)\boldsymbol{\varrho}(t_0)U^\dagger(t,t_0). \tag{B. 2}$$

The evolution operator U is the solution of the Schrödinger equation for the total system T. We know from Section I.1.1 that we can write the solution of the Schrödinger equation when having a time-independent Hamiltonian as an exponential. However here, the total Hamiltonian H_T is obviously time-dependent (only H_0 is assumed constant in time). Nevertheless, a unitary solution still exists in this case [3]: the free system (dynamically governed by H_0) follows unitary dynamics described by $U_0(t,t_0) = e^{-\frac{i}{\hbar}H_0(t-t_0)}$. Thus, we can define the interaction unitary operator U_I as

$$U_I(t,t_0) = U_0^\dagger(t,t_0)U(t,t_0) \tag{B. 3}$$

$$\Leftrightarrow U(t,t_0) = U_0(t,t_0)U_I(t,t_0) \tag{B. 4}$$

If we now try to calculate the average of an operator (in the total space $\mathcal{H}_T$) using the Schrödinger picture, we can find a new interesting relation by injecting the Eq. (B. 4) in the Eq. (B. 2) and then into the following average:

$$\begin{aligned}\langle O(t)\rangle &= \mathrm{Tr}\left[\boldsymbol{\varrho}(t)O(t)\right]\\ &= \mathrm{Tr}\big[\underbrace{U_I(t,t_0)\boldsymbol{\varrho}(t_0)U_I^\dagger(t,t_0)}_{\tilde{\boldsymbol{\varrho}}(t)}\;\underbrace{U_0^\dagger(t,t_0)O(t)U_0(t,t_0)}_{\tilde{O}(t)}\big].\end{aligned}$$

We can thus define in the interaction picture:

$$\tilde{\boldsymbol{\varrho}}(t) = U_I(t,t_0)\boldsymbol{\varrho}(t_0)U_I^\dagger(t,t_0), \tag{B. 5}$$

$$\tilde{O}(t) = U_0^\dagger(t,t_0)O(t)U_0(t,t_0). \tag{B. 6}$$

We can now reverse the Eq. (B. 2) and then use the Eq. (B. 4) to get

$$\boldsymbol{\varrho}(t_0) = U^\dagger(t,t_0)\boldsymbol{\varrho}(t)U(t,t_0) = U_I^\dagger(t,t_0)U_0^\dagger(t,t_0)\boldsymbol{\varrho}(t)U_0(t,t_0)U_I(t,t_0),$$

and, by injecting this into the Eq. (B. 5), we get

$$\tilde{\boldsymbol{\varrho}}(t) = U_0^\dagger(t,t_0)\boldsymbol{\varrho}(t)U_0(t,t_0). \tag{B. 7}$$

This result is obvious because a density matrix is also an operator and thus respects the Eq. (B. 6). In the end, we find the results used in the main part of this document:

$$\boxed{\tilde{\boldsymbol{\varrho}}(t) = U_I(t,t_0)\boldsymbol{\varrho}(t_0)U_I^\dagger(t,t_0) = U_0^\dagger(t,t_0)\boldsymbol{\varrho}(t)U_0(t,t_0)}. \tag{B. 8}$$

By working on the Liouville-von Neumann equation, one can show that by jumping into the interaction picture the dynamics do not depend anymore on the free Hamiltonian H_0:

$$\left(\frac{\mathrm{d}\boldsymbol{\varrho}}{\mathrm{d}t} = -\frac{i}{\hbar}[H_T,\boldsymbol{\varrho}]\right) \Leftrightarrow \left(\frac{\mathrm{d}\tilde{\boldsymbol{\varrho}}}{\mathrm{d}t} = -\frac{i}{\hbar}\left[\tilde{H}_I,\tilde{\boldsymbol{\varrho}}\right]\right), \tag{B. 9}$$

where $\tilde{H}_I = U_0^\dagger(t,t_0)H_IU_0(t,t_0)$ as defined. It is thus usual to work in this representation to get rid of free dynamics terms, and then jump back to the Schrödinger picture to recover the proper dynamics of the system.

Further discussion

We used in Eq. (I.34) the following approximation: $U_I(t+\mathrm{d}t,t) \approx e^{-\frac{i}{\hbar}\tilde{H}_I(t)\mathrm{d}t}$, and we would like to explain it here. As seen in Eq. (B. 5), a state evolves in the interaction picture with a relation depending on U_I, which has been defined as $U_I = U_0^\dagger U$. It is possible to show from this definition that U_I is subject to the following differential equation

$$\frac{\mathrm{d}U_I}{\mathrm{d}t} = -\frac{i}{\hbar}\tilde{H}_I U_I, \tag{B. 10}$$

where we recall that $\tilde{H}_I = U_0^\dagger H_I U_0$ by definition of the interaction picture. This equation is subject to the condition $U_I(t_0,t_0) = \mathbb{I}$. We can now work on the operator $U_I(t+\mathrm{d}t,t_0)$, which can be expressed by a Taylor expansion up to order $\mathrm{d}t$ as

$$U_I(t+\mathrm{d}t,t_0) = U_I(t,t_0) + \frac{\mathrm{d}U_I}{\mathrm{d}t}(t,t_0)\mathrm{d}t + o(\mathrm{d}t). \tag{B. 11}$$

We can then inject Eq. (B. 10) into this expansion to get

$$U_I(t+\mathrm{d}t,t_0) = U_I(t,t_0) - \frac{i}{\hbar}\tilde{H}_I(t,t_0)U_I(t,t_0)\mathrm{d}t + o(\mathrm{d}t). \tag{B. 12}$$

By setting $t = t_0$ and using the condition on unitary operator $U_I(t_0,t_0)$ we obtain

$$U_I(t+\mathrm{d}t,t) = \mathbb{I} - \frac{i}{\hbar}\tilde{H}_I(t)\mathrm{d}t + o(\mathrm{d}t). \tag{B. 13}$$

Note that technically, $\tilde{H}_I(t_0,t_0) = H_I(t_0)$ for the same reason because of U_0. However we are interested into using the expression of $\tilde{H}_I$ in the calculation of Section I.3 and so, we decide to keep the expression in the interaction picture. We can observe that this expansion corresponds to the expansion of the exponential at the same order, indeed

$$e^{-\frac{i}{\hbar}\tilde{H}_I(t)\mathrm{d}t} = \mathbb{I} - \frac{i}{\hbar}\tilde{H}_I(t)\mathrm{d}t + o(\mathrm{d}t), \tag{B. 14}$$

thus proving the expression we used:

$$U_I(t+\mathrm{d}t,t) \approx e^{-\frac{i}{\hbar}\tilde{H}_I(t)\mathrm{d}t}. \tag{B. 15}$$

Appendix C

Some stochastic differential integration

We want here to justify the expression of Eq. (II.12) where we discarded terms proportional to $\mathrm{d}N_t\mathrm{d}t$. We recall that $\mathrm{d}N_t$ is a Poisson increment, meaning that $\mathrm{d}N_t^2 = \mathrm{d}N_t$. Moreover, we work in a case where the only values this increment can have is 0 or 1, with probability $\mathbb{P}[\mathrm{d}N_t = 0] = 1 - \lambda_t\mathrm{d}t \approx 1$ and $\mathbb{P}[\mathrm{d}N_t = 1] = \lambda_t\mathrm{d}t \approx 0$, where $\lambda_t \in \mathbb{R}$ such that $\lambda_t \gg \mathrm{d}t$.

When dealing with stochastic master equation, it is useful to integrate them in order to understand the underlying mathematical background. Note that the real interest of the SME is to integrate it in order to find the expression of a state. We consider the case of a stochastic differential equation, such as

$$\mathrm{d}f = a\ \mathrm{d}t + b\ \mathrm{d}N_t + c\ \mathrm{d}N_t\mathrm{d}t, \tag{C. 1}$$

where $a, b, c \in \mathbb{R}$, they could be functions of time but we prefer to simplify the discussion. We are going to integrate this equation from t_0 to T, to do so we discretize the time as $t_0 < t_1 < \cdots < t_n = T$. We thus express the i-th time as $t_i = t_0 + i\delta t$, in particular when $i = n$: $T = t_0 + n\delta t \Leftrightarrow n = \frac{T-t_0}{\delta t}$. This shows that when $n \to +\infty$, we automatically have $\delta t \to 0^+$. This discretization allows us to define the first integral with measure $\mathrm{d}t$ as

$$\int_{t_0}^{T} a\ \mathrm{d}t = a \lim_{n\to+\infty} \sum_{i=0}^{n-1} (t_{i+1} - t_i) = a \lim_{n\to+\infty} (t_n - t_0) = a(T - t_0), \tag{C. 2}$$

which is simply a Riemann integral. The reason we find a finite quantity is hidden in the fact that we sum infinitely δt while δt is going to 0 at the same time. If we now consider the second integral, that we understand as a Stieltjes integral, we obtain

$$\int_{t_0}^{T} b\ \mathrm{d}N_t = b \lim_{n\to+\infty} \sum_{i=0}^{n-1} [N(t_{i+1}, t_0) - N(t_i, t_0)] = b \sum_{j=0}^{k-1} 1 = bk, \tag{C. 3}$$

where $N(t_{i+1}, t_0) - N(t_i, t_0) = 0$ most of the time and 1 only k times, according to the probability law followed by the Poisson increment. This result, when applied to the third integral, gives

$$\int_{t_0}^{T} c\ \mathrm{d}N_t\mathrm{d}t = c \sum_{j=0}^{k-1} \mathrm{d}t = ck\mathrm{d}t \approx 0, \tag{C. 4}$$

in the Stieltjes definition of the integral, which explains why we can neglect $\mathrm{d}N_t\mathrm{d}t$ in the photo-detection SME.

Appendix D

Homodyne detection in quantum optics

As explained in Section II.4, homodyne detection creates a continuous signal containing information on the quadrature of the physical system of interest. We use the notations of Figure 2, i.e. the system field is labelled by the system annihilation operator $\hat{a}$ (and is now an input of the homodyne detection process), also we use another input signal which is a coherent field labelled by its annihilation operator $\hat{b}$. Moreover, we use a 50:50 beam-splitter to form two output fields labelled $\hat{c}$ and $\hat{d}$ which are then measured by two photo-detector, as depicted in the aforementioned figure. The quadrature of the system is defined by

$$\hat{a}_\theta = \frac{\hat{a}^\dagger e^{i\theta} + \hat{a} e^{-i\theta}}{\sqrt{2}}. \tag{D. 1}$$

Quantum optics [18] assures that, after the beam-splitter, the output fields are

$$\hat{c} = \frac{\hat{a} + i\hat{b}}{\sqrt{2}}, \qquad \hat{d} = \frac{\hat{b} + i\hat{a}}{\sqrt{2}}. \tag{D. 2}$$

These relations hold when the beam-splitter is 50:50. Hence, this exact process is called "balanced" homodyne detection. We explained in Section II.3 that the photo-detector signal is proportional to the mean value of the occupation number. Thus, $I_c \propto \left\langle \hat{c}^\dagger \hat{c} \right\rangle$ and $I_d \propto \left\langle \hat{d}^\dagger \hat{d} \right\rangle$. Therefore, the difference of photo-currents is

$$I_d - I_c \propto i \left\langle \hat{a}^\dagger \otimes \hat{b} - \hat{a} \otimes \hat{b}^\dagger \right\rangle. \tag{D. 3}$$

As explained, the field input labelled by $\hat{b}$ is a coherent field. Let us note it as $|\beta\rangle$, this state respects: $\hat{b}\,|\beta\rangle = \beta\,|\beta\rangle$, where $\beta = |\beta| e^{i\phi} \in \mathbb{C}$. If we denote as $|\psi\rangle$ the input state of the system, then the total input state is $|\psi_{\text{in}}\rangle = |\psi\rangle \otimes |\beta\rangle$. We can therefore explicit the average in Eq. (D. 3) as

$$I_d - I_c \propto i|\beta| \left\langle \psi \right| (\hat{a}^\dagger e^{i\phi} - \hat{a} e^{-i\phi}) \left| \psi \right\rangle = |\beta| \left\langle \psi \right| (\hat{a}^\dagger e^{i\theta} + \hat{a} e^{-i\theta}) \left| \psi \right\rangle, \tag{D. 4}$$

where we introduced $\theta = \phi - \pi$. Therefore, we have briefly shown that homodyne detection signal is proportional to the quadrature of the system $\hat{a}_\theta$.

Bibliography

[1] Richard P. Feynman, Robert B. Leighton, and Matthew Sands. *The Feynman Lectures on Physics, Vol. III, Chapter 1: Quantum Behavior*. California Institute of Technology, accessed 2026-04-02. 1965. URL: https://www.feynmanlectures.caltech.edu/III_01.html.

[2] Claude Cohen-Tannoudji, Bernard Diu, and Franck Laloë. *Mécanique quantique*. fr. Companyédition CNRS/EDP Sciences, Sept. 2018.

[3] Heinz-Peter Breuer and Francesco Petruccione. *The Theory of Open Quantum Systems*. Oxford University PressOxford, Jan. 2007. ISBN: 9780191706349. DOI: 10.1093/acprof:oso/9780199213900.001.0001. URL: http://dx.doi.org/10.1093/acprof:oso/9780199213900.001.0001.

[4] Kurt Jacobs. *Quantum Measurement Theory and its Applications*. en. Cambridge, England: Cambridge University Press, Aug. 2014.

[5] Francesco Albarelli and Marco G. Genoni. "A pedagogical introduction to continuously monitored quantum systems and measurement-based feedback". In: *Physics Letters A* 494 (Jan. 2024), p. 129260. ISSN: 0375-9601. DOI: 10.1016/j.physleta.2023.129260. URL: http://dx.doi.org/10.1016/j.physleta.2023.129260.

[6] Howard Mark Wiseman. "Quantum trajectories and feedback". PhD thesis. University of Queensland, 1994.

[7] Howard Mark Wiseman. "Quantum trajectories and quantum measurement theory". In: *Quantum and Semiclassical Optics: Journal of the European Optical Society Part B* 8.1 (Feb. 1996), pp. 205–222. ISSN: 1361-6625. DOI: 10.1088/1355-5111/8/1/015. URL: http://dx.doi.org/10.1088/1355-5111/8/1/015.

[8] Alexander S. Holevo. *Probabilistic and statistical aspects of quantum theory*. en. Publications of the Scuola Normale Superiore. Pisa, Italy: Scuola Normale Superiore, Mar. 2011.

[9] Michael A. Nielsen and Isaac L. Chuang. *Quantum Computation and Quantum Information*. Cambridge, England: Cambridge University Press, Dec. 2010.

[10] János A. Bergou, Mark Hillery, and Mark Saffman. *Quantum Information Processing: Theory and Implementation*. Springer International Publishing, 2021. ISBN: 9783030754365. DOI: 10.1007/978-3-030-75436-5. URL: http://dx.doi.org/10.1007/978-3-030-75436-5.

[11] C. W. Gardiner and M. J. Collett. "Input and output in damped quantum systems: Quantum stochastic differential equations and the master equation". In: *Phys. Rev. A* 31 (6 June 1985), pp. 3761–3774. DOI: 10.1103/PhysRevA.31.3761. URL: https://link.aps.org/doi/10.1103/PhysRevA.31.3761.

[12] Jonas Lammers. "State preparation and verification in continuously measured quantum systems". en. In: (2018). DOI: 10.15488/3854. URL: https://www.repo.uni-hannover.de/handle/123456789/3888.

[13] Peter Zoller and C. W. Gardiner. *Quantum Noise in Quantum Optics: the Stochastic Schrödinger Equation*. 1997. arXiv: quant-ph/9702030 [quant-ph]. URL: https://arxiv.org/abs/quant-ph/9702030.

[14] Howard Carmichael. *An Open Systems Approach to Quantum Optics: Lectures Presented at the Université Libre de Bruxelles October 28 to November 4, 1991*. Springer Berlin Heidelberg, 1993. ISBN: 9783540476207. DOI: 10.1007/978-3-540-47620-7. URL: http://dx.doi.org/10.1007/978-3-540-47620-7.

[15] Howard M Wiseman and Gerard J Milburn. *Quantum Measurement and Control*. en. Cambridge, England: Cambridge University Press, Nov. 2009.

[16] Kurt Jacobs and Daniel A. Steck. "A straightforward introduction to continuous quantum measurement". In: *Contemporary Physics* 47.5 (2006), pp. 279–303. ISSN: 1366-5812. DOI: 10.1080/00107510601101934. URL: http://dx.doi.org/10.1080/00107510601101934.

[17] Pierre Rouchon. "A tutorial introduction to quantum stochastic master equations based on the qubit/photon system". In: *Annual Reviews in Control* 54 (2022), pp. 252–261. ISSN: 1367-5788. DOI: 10.1016/j.arcontrol.2022.09.006. URL: http://dx.doi.org/10.1016/j.arcontrol.2022.09.006.

[18] Christopher C. Gerry and Peter L. Knight. *Introductory Quantum Optics*. Cambridge University Press, Nov. 2023. ISBN: 9781107653948. DOI: 10.1017/9781139151207. URL: http://dx.doi.org/10.1017/9781139151207.

[19] Stephen M. Barnett and Paul M. Radmore. *Methods in Theoretical Quantum Optics*. Oxford University PressOxford, Nov. 2002. ISBN: 9780191714245. DOI: 10.1093/acprof:oso/9780198563617.001.0001. URL: http://dx.doi.org/10.1093/acprof:oso/9780198563617.001.0001.